The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Abstract

Objective. We develop a mesoscale model of temporal-interference (TI) stimulation with three stages in series: membrane access, nonlinear detection of the difference frequency absent from the field, and circuit amplification. **Approach.** We represent access by an effective membrane time and analyze detection and amplification in a Jansen–Rit column, an exact mean field of quadratic integrate-and-fire (QIF) neurons, and its spiking network. We compare spatial maps in one head model and quantify conditional kHz access budgets. **Main results.** For weak fields, a static quadratic detector generates a beat proportional to the carrier-polarization product and a signed, direct-current-like (DC-like) steady source proportional to their summed squares. Transcranial alternating-current (tACS) and direct-current (tDCS) stimulation sources are linear to leading order. At the pinned Jansen–Rit operating point, 0.1–0.3 mV per carrier gives TI sources approximately 2–5% of direct sources at equal per-carrier and direct-input polarizations. Dynamic QIF detection crosses from static curvature to positive-slope rectification at fast carriers. Changing the current ratio moves the hippocampal envelope and mean-square centroids oppositely but only rescales the beat map at fixed state. Near a Hopf bifurcation, resonant gain grows inversely with modal damping and is shared with tACS. In the pinned Jansen–Rit column, coupling changes gain independently of detection; amplification appears mainly in beat-locked population-rate modulation, with little mean-rate change. At stronger inputs, the steady source alone drives the circuit across its gamma threshold, as does an unmodulated carrier matched in root-mean-square (RMS) input. **Significance.** Weak-field beat dose follows the polarization product rather than the envelope minimum. The steady source can act at depth; its sign depends on detector state and dynamics, not carrier polarity. For uniform scalar coupling and detection in homogeneous conductivity, its magnitude has no strict interior maximum. Polarization measurements and amplitude and frequency sweeps test the stages; physiological kHz coupling remains uncalibrated.

Keywords: temporal interference; non-invasive neuromodulation; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; criticality.

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1 Introduction

Temporal-interference (TI) stimulation was proposed to reach deep brain targets while reducing stimulation of surrounding tissue [1]. It applies two sinusoids at f_1 and $f_2 = f_1 + \Delta f$ through separate electrode pairs, typically with $f_1 \sim 1$–2 kHz and $\Delta f \sim 1$–100 Hz. Where the fields overlap, their superposition is an amplitude-modulated (AM) signal with an envelope at Δf. Electrode currents can steer the region of largest envelope modulation. Whether this also localizes the neural response depends on how neurons transduce the waveform.

The difference frequency is absent from the input itself:

$$s(t) = \varepsilon_1 \cos(\omega_1 t) + \varepsilon_2 \cos(\omega_2 t), \qquad \omega_{1,2} = 2\pi f_{1,2}, \quad \Omega = \omega_2 - \omega_1 = 2\pi\,\Delta f. \tag{1}$$

Here $s(t)$ is the local membrane polarization, t is time, and $\varepsilon_{1,2}$ are its tone amplitudes in mV, produced by coupling of the applied field to the membrane; Ω is the beat angular frequency, and $f_c = (f_1 + f_2)/2$ is the carrier frequency. The spectrum contains only the two applied lines. A linear, time-invariant system can attenuate or phase-shift them but cannot create a new one. Passive membrane filtering therefore cannot demodulate TI [2]; a nonlinear element must mix the carriers into the low-frequency band, as in AM reception [3].

The mechanism has three physical requirements. *Access* is the conversion of the extracellular field into local membrane polarization. *Detection*, or demodulation, is the nonlinear generation of the difference-frequency component from those polarizations. *Amplification* is the response of the neuron and its recurrent circuit to the generated component. Somatic charging attenuates kHz fields, with the response also depending on cable geometry and field coupling [4, 5]. Esmaeilpour et al. [6] already modeled TI sensitivity using a short, axon-like membrane time. Cohen et al.'s conducting periaxonal space separates nodal and internodal voltage dynamics and supports saltatory conduction [7]; nodes and the sodium-channel-rich axon initial segment (AIS) are candidate access sites [8]. We represent this route by a short effective membrane time, whose field-to-polarization transfer requires calibration.

Detection requires that this polarization encounter a nonlinearity. Mirzakhalili et al. [2] demonstrated subthreshold nodal mixing in a modified McIntyre–Richardson–Grill (MRG) myelinated-axon model with Hodgkin–Huxley ion-channel dynamics. Cortical-neuron models also show axonal excitation and conduction block under stronger fields [9]. Luff et al. [10] directly observed subthreshold difference-frequency oscillations in individual neurons, including carriers near 5 kHz. Mixing persisted during synaptic blockade but was suppressed by blocking tetrodotoxin-sensitive sodium channels. This establishes cellular detection without locating its compartment. Smooth cellular responses admit a population-level second-order expansion; a static rate curve supplies it through curvature.

Amplification determines whether the generated beat appreciably changes activity. Intrinsic currents, including the hyperpolarization-activated current I_h and M-type potassium current I_M, can tune a single neuron [11], as in the cell- and state-dependent hippocampal TI responses of de Palma Aristides et al. [12]. Recurrent excitatory–inhibitory circuits supply another resonance [13, 14, 15]. In slices, pyramidal and parvalbumin (PV) neurons respond differently, and most pyramidal TI responses vanish under synaptic blockade [16]; simulations also find network mediation [17, 6]. Reviews summarize these mechanisms [18, 19]. At human-comparable fields, primate recordings show weaker effects than direct low-frequency stimulation, expressed mainly in spike timing rather than mean rate [20].

We focus on weak stimulation-induced changes around ongoing activity, which may already include spiking. Direct carrier-evoked spiking and conduction block [21, 22, 23] lie outside this expansion; the finite-amplitude network examples are identified separately.

The framework in Fig. 1 combines cellular mixing with recurrent gain, extending models of endogenous cross-frequency coupling [24, 25]. Cassarà et al. already identified the fixed-pattern

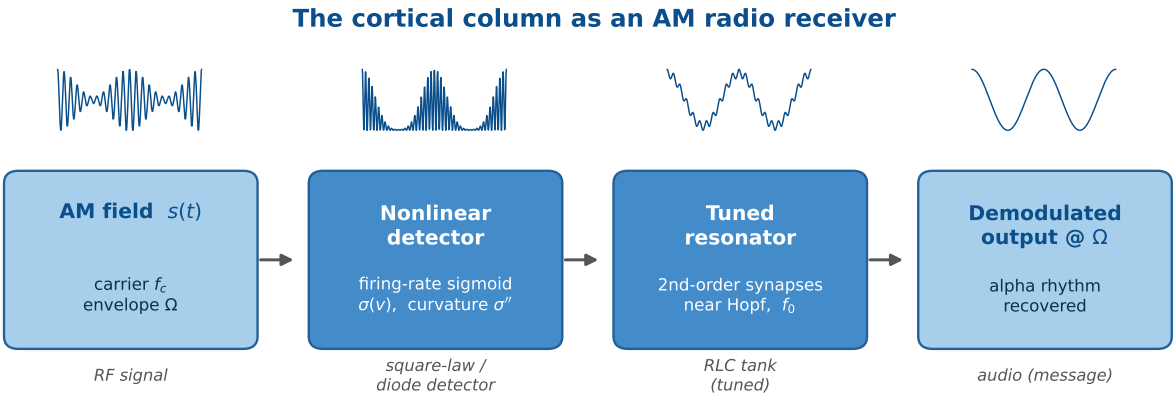


Figure 1: The cortical column as a tuned receiver. Access converts the applied field into the local polarization entering the detector. The nonlinear detector generates the beat at angular frequency Ω from a carrier near f_c; a recurrent network amplifies it near its natural frequency f_0. The diagram shows the static-sigmoid realization. In a conventional AM receiver, the tuned circuit selects the carrier before detection [3]; here the resonator selects the low-frequency output.

consequence of second-order mixing [26]. Here we derive steady and beat sources together, separate detector curvature from recurrent gain at a pinned Jansen–Rit (JR) operating point, and derive the static-to-dynamic crossover in an exact QIF mean field. A head-model illustration compares the resulting spatial maps, while equal-RMS carrier controls distinguish beat-dependent modulation from induced oscillations. These results specify a mechanism and its conditional input budget; the field-to-detector transfer at physiological kHz fields remains uncalibrated.

2 Theory

2.1 A population rate curve supplies the nonlinearity

A neural mass model describes a population through averaged variables [27, 28]. In the Jansen–Rit (JR) family, a linear synaptic filter converts presynaptic firing rate $r(t)$ into postsynaptic potential $u_{\mathrm{PSP}}(t)$:

$$\ddot{u}_{\mathrm{PSP}} + 2a\dot{u}_{\mathrm{PSP}} + a^2 u_{\mathrm{PSP}} = Aa\, r(t),$$

with inverse synaptic time a (s^{-1}) and PSP gain A (mV); inhibitory synapses use b and B in the same respective units. Summed postsynaptic potentials give the mean input v, which a static transfer curve converts to firing rate [29],

$$\sigma(v) = \frac{2e_0}{1 + e^{\rho(v_0 - v)}}. \tag{2}$$

Here $2e_0$ is the maximum rate, v_0 the inflection voltage, and ρ the inverse voltage scale (s^{-1}, mV, and mV^{-1}, respectively). Primes denote derivatives with respect to voltage. Nonzero curvature makes this curve a mixer, while its slope sets the local gain of the recurrent circuit.

The JR column couples pyramidal, excitatory-interneuron, and inhibitory-interneuron populations, with field input to the pyramidal sigmoid and local field potential (LFP) proxy $v = y_1 - y_2$, where y_1 and y_2 are the excitatory and inhibitory postsynaptic potentials onto the pyramidal population. Its external input rate is p (s^{-1}), and C is the dimensionless connectivity scale.

Figure 2 locates the field input, nonlinear detector, and recurrent amplifier in the JR and QIF circuits.

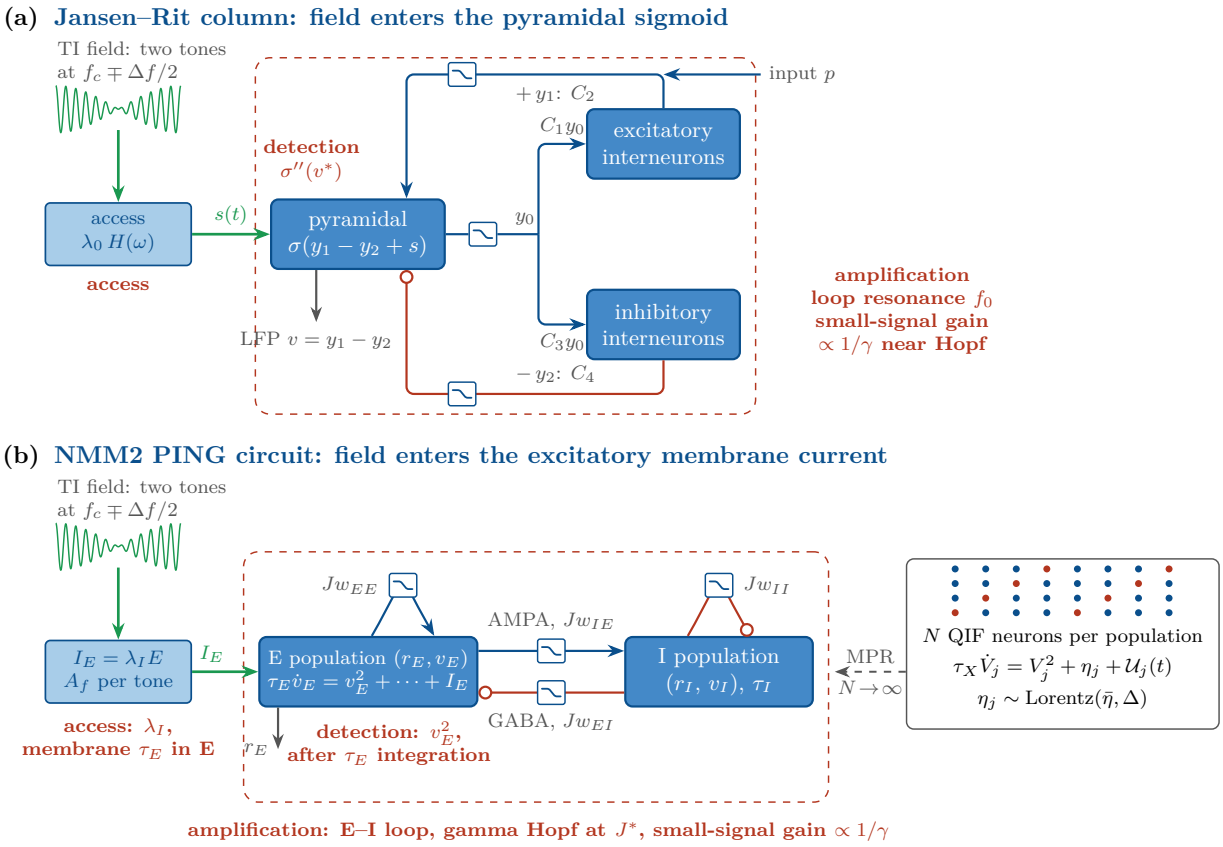


Figure 2: Access, detection, and amplification in the two circuits. *(a)* JR: access $\lambda_0 H(\omega)$ converts the field to polarization s, and sigmoid curvature mixes the tones. *(b)* NMM2 PING: current $I_E = \lambda_I E$ enters the excitatory membrane; v_E^2 mixes its voltage excursions. The inset shows the QIF neurons whose infinite-population limit gives this exact mean field. Low-pass symbols mark second-order synapses (h_e, h_i in a; AMPA and GABA with times τ_a, τ_g in b). They suppress carriers more strongly than the generated beat and transmit the steady shift. Dashed frames enclose recurrent amplifiers, with small-signal gain proportional to $1/\gamma$ near the relevant Hopf. Blue: excitation; red: inhibition; green: field input.

2.2 From applied field to a squared-envelope beat component

We neglect electromagnetic propagation and induction in the applied field $E(t)$ (V/m), while retaining the membrane's finite charging response. Let $s(t)$ be the effective polarization (mV) expressed in the sigmoid's input coordinate after cellular access; the static population model reads

$$r(t) = \sigma(v(t) + s(t)), \qquad E(t) = E_1 \cos \omega_1 t + E_2 \cos \omega_2 t. \tag{3}$$

Soma, dendrites, AIS, nodes, and terminals have different couplings and response times. We represent their contributions as parallel inputs. For nearby carriers centered at $\omega_c = 2\pi f_c$,

$$s(t) \simeq \sum_k \boldsymbol{\lambda}_f^k(\omega_c) \cdot \mathbf{E}(t), \qquad \boldsymbol{\lambda}_f^k(\omega) = \boldsymbol{\lambda}_0^k \, |H_k(\omega)|, \tag{4}$$

where $\boldsymbol{\lambda}_0^k$ is the static polarization vector of compartment k (a length), and H_k its access filter [27, 30]. Equation (4) uses amplitude gains and assumes aligned compartment phases; a general sum requires the complex transfers, including their relative phases. The supplied polarization includes the access response.

For a dominant entry point, define $\lambda_f(\omega) = \lambda_0 |H(\omega)|$. Then $\varepsilon_j = \lambda_f(\omega_j) E_j$ denotes the polarization reaching the nonlinearity. We first hold this local polarization fixed to isolate mixing and recurrent amplification. At fixed applied field, its frequency dependence follows H. Fast axonal

compartments motivate the direct-coupling approximation $\lambda_f \approx \lambda_0$. For coupling at the AIS, the additive input s represents a change in spike-initiation drive: $\sigma(v+s)$ is the same sigmoid with its effective threshold shifted from v_0 to $v_0 - s$. No backward transmission to the soma is required. Converting local AIS polarization into s, and determining the compartment's response time, require calibration in the preparation. Fast access alone does not establish that the static sigmoid describes detection at the carrier frequency.

Expanding σ about a field-free operating voltage v^* (mV) (a Volterra/Taylor expansion, as in [25]) gives $\sigma(v^* + s) \approx \sigma(v^*) + \sigma'(v^*)\, s + \frac{1}{2}\sigma''(v^*)\, s^2 + \dots$ The truncation requires $|s| \ll 1/\rho \approx 1.8$ mV and a nonzero second derivative for quadratic mixing to be leading. Larger excursions require the full transfer curve. The linear term preserves the two input frequencies. The quadratic cross term generates their difference through the identity $2\cos\omega_1 t \cos\omega_2 t = \cos\Omega t + \cos(\omega_1 + \omega_2)t$. Low-pass synaptic filtering suppresses the fast terms after mixing; membrane filtering, when retained, already shapes the carrier polarization. We isolate the slow source by a carrier average, before applying its frequency-dependent network gain. The fast $\cos^2\omega_j t$ terms average to $\frac{1}{2}$ and the sum-frequency term to zero:

$$\left\langle \sigma(v^* + s) \right\rangle_c = \sigma(v^*) + \tfrac{1}{4}\sigma''(v^*)\,(\varepsilon_1^2 + \varepsilon_2^2) + \tfrac{1}{2}\sigma''(v^*)\,\varepsilon_1\varepsilon_2 \cos\Omega t + O(\varepsilon^4), \tag{5}$$

so a line appears at exactly the envelope frequency Ω, with amplitude

$$Y_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon_1\varepsilon_2, \tag{6}$$

Here Y_Ω is the signed rate amplitude before recurrent feedback, in s^{-1}; ε denotes a common small amplitude scale, and $\simeq$ means to leading order in the polarizations, here and in what follows. The same expansion leaves a constant (DC) term, $\frac{1}{4}\sigma''(\varepsilon_1^2 + \varepsilon_2^2)$, which any read-out of the mean rate must include.

For a dynamic cellular response with synaptic input held fixed, define the effective second-order coefficient $\chi^{(2)}$ by the complex beat amplitude Y_Ω,

$$Y_\Omega \simeq \chi^{(2)}(\omega_2, -\omega_1)\,\varepsilon_2\varepsilon_1, \qquad r_\Omega(t) = \mathrm{Re}\left[Y_\Omega e^{i\Omega t}\right]. \tag{7}$$

Here $r_\Omega(t)$ is the generated rate component before recurrent feedback. Thus $\chi^{(2)}$ has units $\mathrm{s}^{-1}\mathrm{mV}^{-2}$ when its input is polarization and its output is physical rate. With this cosine-amplitude convention, the memoryless limit is $\chi^{(2)} = \sigma''(v^*)/2$. Cellular processes contribute to this coefficient according to their state and response times. Weak input permits a second-order expansion, but a response that depends on input history retains frequency-dependent amplitude and phase. If symmetry or cancellation makes this coefficient zero, higher orders determine the leading beat.

For the static detector, Eq. (6) predicts quadratic amplitude scaling, a source that vanishes at the sigmoid inflection, and carrier independence at fixed post-access polarization. At fixed applied field, $\varepsilon = \lambda_f E_0$ instead inherits the access filter, so attenuation raises the field needed for a given source, consistent with the carrier-dependent thresholds in [21, 22].

2.3 Envelope oscillations and spatial weighting

Envelope following and quadratic detection assign different spatial doses to the same two-tone field. Write $\mathbf{x}$ for position and $\varepsilon_j(\mathbf{x})$ for the signed local polarization of tone j, projected along a fixed direction. These amplitudes vary in space and are held fixed only during the temporal averages at one location. The nonnegative envelope is

$$\mathcal{A}(\mathbf{x}, t) = \sqrt{\varepsilon_1(\mathbf{x})^2 + \varepsilon_2(\mathbf{x})^2 + 2\varepsilon_1(\mathbf{x})\varepsilon_2(\mathbf{x})\cos\Omega t}.$$

Carrier averaging of the squared signal gives

$$\mathrm{LPF}[s^2(t)] = \tfrac{1}{2}\mathcal{A}^2(t) = \frac{\varepsilon_1^2 + \varepsilon_2^2}{2} + \varepsilon_1\varepsilon_2\cos\Omega t, \tag{8}$$

with spatial arguments suppressed at a fixed location. Thus square-law detection recovers $\mathcal{A}^2$, which is sinusoidal at Ω, whereas the envelope itself is not: at a matched site, $\mathcal{A}(t) = 2\varepsilon|\cos(\Omega t/2)|$, a rectified cosine. The conventional TI map describes its excursion, $\mathcal{M}[y] \equiv \max_t y - \min_t y$, along the chosen direction [1]. Table 1 compares this envelope with the squared-envelope signal; the overbar denotes a beat-period mean. The geometric maps precede detector gains: an ideal envelope follower has response $c_1\mathcal{A}$ for a specified gain c_1. A smooth transfer g with nonzero curvature gives the weak response $g''(v^*)\mathcal{A}^2/4$; for the JR column, $g = \sigma$. The last row carries these gains into the signed steady source. In particular, the nonnegative mean-square polarization $\mathcal{I}$ alone does not specify excitation or suppression. Envelope following is an additional response assumption; the conventional envelope-excursion map alone does not specify a neuron's mean output.

	Envelope following	Quadratic detection
Slow signal	$\mathcal{A}(\mathbf{x}, t)$	$\frac{1}{2}\mathcal{A}(\mathbf{x}, t)^2$
Mean before gain	$\overline{\mathcal{A}}(\mathbf{x})$	$\mathcal{I}(\mathbf{x}) = \dfrac{\varepsilon_1^2 + \varepsilon_2^2}{2}$
Modulation excursion	$2\min(\lvert\varepsilon_1\rvert, \lvert\varepsilon_2\rvert)$	$2\lvert\varepsilon_1\varepsilon_2\rvert$
Units before detector gain	mV	mV^2
Signed steady source (output units)	$c_1\overline{\mathcal{A}}$	$Y_{\mathrm{DC}} = \frac{1}{2}g''(v^*)\mathcal{I}$

Table 1: Envelope and quadratic maps at the same position $\mathbf{x}$. The first four rows give geometric quantities before detector gain; the last gives the steady source after gain, before recurrent feedback. Its sign depends on the detector. Averaging the slow signal over a beat period gives its mean. Excursion means maximum minus minimum, not Fourier amplitude.

The excursion maps in Table 1 are related pointwise by

$$\mathcal{M}[\tfrac{1}{2}\mathcal{A}^2] = \max(|\varepsilon_1|, |\varepsilon_2|)\, \mathcal{M}[\mathcal{A}]. \tag{9}$$

The dominant local polarization therefore reweights the envelope map. Both maps can have a central modulation maximum; the quadratic profile in Fig. S13a is broader. If the dominant carrier is larger superficially, this reweighting degrades deep-to-superficial contrast. For $(\varepsilon_1, \varepsilon_2) = (10, 1)$ superficially and $(0.5, 0.5)$ at depth, the envelope excursion favors the superficial point by $2\times$ and the quadratic excursion by $40\times$. A sufficiently weak second carrier can instead suppress the superficial product. These examples illustrate the algebra; realistic contrast requires the actual vector fields.

With one fixed electrode pair per carrier and fixed neural state, changing the current ratio rescales the leading quadratic beat map without moving its maxima. For fixed frequencies, linear access, and fixed projection directions, write $\varepsilon_j(\mathbf{x}) = I_j^{\mathrm{pair}}\mathcal{P}_j(\mathbf{x})$, where $I_j^{\mathrm{pair}} > 0$ is the peak pair current (mA) and $\mathcal{P}_j$ is the signed local polarization per unit current (mV/mA). Then

$$\mathcal{M}[\tfrac{1}{2}\mathcal{A}^2] = 2I_1^{\mathrm{pair}} I_2^{\mathrm{pair}} |\mathcal{P}_1\mathcal{P}_2|.$$

The normalized map and its ratios between fixed target and reference regions are unchanged, as Cassarà et al. noted for second-order mixing [26]. Fixed detector coefficients and linear network propagation preserve this common current factor, including for dynamic mixing and spatially heterogeneous gains. Steering within this regime requires changing at least one carrier's field shape, for example by redistributing current among several electrodes at that frequency. The mean-square polarization map, however, weights $\mathcal{P}_1^2$ and $\mathcal{P}_2^2$ separately by the squared currents. It changes with the ratio; after signed detection, the resulting steady source can indirectly reshape the beat response by shifting local operating points. Threshold effects can also break the quadratic

prediction; the conventional 2 min steering law applies when the detector follows the envelope. An interior beat maximum remains compatible with the fixed-state product law.

The local response weights the carrier product by the detector curvature and network susceptibility (Eq. (12)). Field overlap supplies the source; local physiology sets its sign and amplification. Resonance can compensate for an unfavorable field product but requires a nonzero source.

The scalar maps use a common fixed projection, $\varepsilon_j = \boldsymbol{\lambda} \cdot \mathbf{E}_j$; the beat changes phase when either projection changes sign. More generally, let a_j^k be the complex polarization of carrier j in compartment k. A weak dynamic detector has a bilinear beat source $Y_\Omega = \sum_{kl} B_{kl}(\omega_2, -\omega_1) a_2^k \overline{a_1^l}$, where B_{kl} are fixed response coefficients and the overbar means complex conjugation. A single scalar input is a rank-one specialization. Linear access gives $a_j^k = I_j^{\mathrm{pair}} b_j^k$, with b_j^k the polarization per unit current, so the common factor $I_1^{\mathrm{pair}} I_2^{\mathrm{pair}}$ and its fixed spatial pattern survive without a universal projection vector.

2.4 Detector laws and harmonic signatures

The detector's transduction $T(\mathcal{A})$ determines amplitude scaling, harmonic content, and spatial weighting.

For a narrowband field $s(t) = \mathcal{A}(t)\cos\omega_c t$ passing through a memoryless nonlinearity g, the low-frequency (baseband) output Y_{LF} is the average of g over one carrier cycle, with carrier phase θ:

$$Y_{\mathrm{LF}}(\mathcal{A}) = (2\pi)^{-1} \int_0^{2\pi} g(v^* + \mathcal{A}\cos\theta)\, \mathrm{d}\theta.$$

Odd powers of $\mathcal{A}$ average away, leaving only even ones:

$$Y_{\mathrm{LF}}(\mathcal{A}) = g(v^*) + \tfrac{1}{4} g''(v^*)\mathcal{A}^2 + \tfrac{1}{64} g^{(4)}(v^*)\mathcal{A}^4 + \cdots \tag{10}$$

Thus any smooth memoryless response with $g''(v^*) \neq 0$ is quadratic at leading order. Envelope following requires a different detector: full- and half-wave rectifiers give $(2/\pi)\mathcal{A}$ and $\mathcal{A}/\pi$, respectively, after low-pass filtering, and a diode-and-capacitor detector follows the peak. A step detector with offset $\delta > 0$ instead gives $\pi^{-1} \arccos(\delta/\mathcal{A})$ above threshold and zero below it. Channel rectification, threshold crossing, and release events are candidate biological implementations of different laws.

At matched carrier amplitudes, the rectified-cosine envelope has a second-to-first harmonic ratio of 1/5, whereas the quadratic output contains only the steady term and the fundamental. The full harmonic series and an inversion for mixed linear/quadratic detectors are given in the supplement (§S2.2).

A smooth detector generates a second harmonic beyond quadratic order, $Y_{2\Omega}/Y_\Omega \sim (c_4/c_2)\varepsilon^2$, where $c_2 = g''(v^*)/4$ and $c_4 = g^{(4)}(v^*)/64$ are the coefficients in Eq. (10). The square-law signature is a harmonic ratio tending to zero with input strength. The recurrent network also multiplies each harmonic by its frequency-dependent gain G, $\mathcal{H}_{\mathrm{meas}} = \mathcal{H}_{\mathrm{det}}\, |G(2\Omega)/G(\Omega)|$. Near resonance, harmonic suppression can make an envelope follower appear square-law.

Amplitude dependence distinguishes these cases: envelope following is linear with harmonic ratio 1/5; weak quadratic detection is quadratic with a ratio vanishing as ε^2; a threshold detector is silent for $2\varepsilon \leq \delta$ and has an amplitude-dependent ratio after onset. A single harmonic ratio is insufficient.

2.5 The steady source has a state-dependent sign

Mean-square polarization is a nonnegative geometric map. Detection converts it into a signed, DC-like steady source, and the circuit converts that source into a mean response. *DC-like* refers to the

detected source, not to a DC component in the applied field. For the scalar detector with uniform conductivity, access, orientation, and detection, its magnitude has no strict interior maximum; the ideal envelope-following mean obeys the same restriction. The envelope mean is

$$\overline{\mathcal{A}}(\mathbf{x}) = \frac{1}{2\pi}\int_0^{2\pi} \sqrt{\varepsilon_1(\mathbf{x})^2 + \varepsilon_2(\mathbf{x})^2 + 2\varepsilon_1(\mathbf{x})\varepsilon_2(\mathbf{x})\cos\vartheta}\,\mathrm{d}\vartheta,$$

where ϑ is beat phase. At a matched site with $\varepsilon_1 = \varepsilon_2 = \varepsilon > 0$, the envelope and quadratic means are $4\varepsilon/\pi$ and ε^2, respectively. These means become detector outputs after multiplication by their gains; the quadratic response change takes the sign of g''. Even one carrier gives either ideal detector a mean output, but no beat modulation; adaptation can alter the physiological mean response.

For the JR quadratic detector, the local rate source in Eq. (5) is

$$Y_{\mathrm{DC}}(\mathbf{x},\mathbf{n}) = \frac{\sigma''(v^*_{\mathbf{x}})}{2}\,\mathcal{I}(\mathbf{x},\mathbf{n}), \qquad \mathcal{I}(\mathbf{x},\mathbf{n}) = \frac{\varepsilon_1(\mathbf{x},\mathbf{n})^2 + \varepsilon_2(\mathbf{x},\mathbf{n})^2}{2} \geq 0.$$

Here $v^*_{\mathbf{x}}$ is the baseline voltage at position $\mathbf{x}$ and $\mathbf{n}$ is the unit cell orientation used for both signed polarizations. The source Y_{DC} is positive on the convex part of the rate curve, negative on its concave part, and zero to this order at the inflection. Reversing either carrier's polarity or $\mathbf{n}$ leaves the squares unchanged; other directions change their magnitudes, while the sign remains set by the fixed curvature. Figure S13c shows both signs using one positive normalization. At the pinned JR operating point $v^* = 8.30$ mV, $\sigma'' < 0$, so the static source lowers the local rate. This curvature sign rule belongs to the static detector. Dynamic QIF rectification instead follows a positive slope at fast carriers (§2.10); the final mean response also depends on recurrence.

Carrier averaging retains the cross term as the beat; averaging over many beat cycles removes it, leaving a sum of intensities without cross-frequency cancellation. At the detector output, where $Y_{\mathrm{DC}} \neq 0$, $|Y_\Omega/Y_{\mathrm{DC}}| = 2|\varepsilon_1\varepsilon_2|/(\varepsilon_1^2+\varepsilon_2^2) \leq 1$, with equality at matched tone magnitudes. A common access factor cancels from this local ratio. In the polarization mean-square map $\mathcal{I} = (\varepsilon_1^2 + \varepsilon_2^2)/2$, the example above gives 50.5 superficially and 0.25 at depth, a 202-fold contrast; the beat/DC ratios are 0.20 and 1, respectively.

The geometric restriction follows in a homogeneous, isotropic, source-free conductor. Writing ϕ_j for the potential of tone j, the potentials ϕ_j, each component of $\mathbf{E}_j = -\nabla\phi_j$, and the signed polarizations $\varepsilon_j = \boldsymbol{\lambda}\cdot\mathbf{E}_j$ for spatially fixed coupling $\boldsymbol{\lambda}$ are harmonic. Thus $\nabla^2\mathcal{I} = \sum_j |\nabla\varepsilon_j|^2 \geq 0$: the intensity map is subharmonic and has no strict interior maximum. A montage can shape each carrier field, but the maximum of this map lies on the region's boundary; conductivity interfaces are such boundaries. For a spatially constant nonzero detector coefficient, this bounds the magnitude of Y_{DC}. A negative coefficient makes the source most negative at the boundary. Spatially varying coefficients require a separately weighted map. A general multicompartment DC quadratic form need not be sign-definite and is not covered by this scalar boundary result.

The envelope mean is also subharmonic under these assumptions: at each beat phase it is the norm of a vector of harmonic functions, and phase averaging preserves subharmonicity (supplement, §S3.1). This does not prohibit a beat maximum: the product instead obeys $\nabla^2(\varepsilon_1\varepsilon_2) = 2\nabla\varepsilon_1\cdot\nabla\varepsilon_2$, whose sign is unrestricted. Within the homogeneous model, interior selectivity must therefore come from the oscillatory response rather than from either mean map.

Spatially varying access, orientation, or curvature can change the signed-source map and the resulting physiological mean-response map. The intensity result does not extend to these weighted responses. For isotropic orientations with orientation-independent access and curvature, orthogonal fields cancel the averaged beat but retain the steady intensity (supplement, §S3.2). At equal local RMS polarization, an unmodulated carrier gives the same mean shift at second order when its DC mixing coefficient equals those of both TI carriers. For a dynamic detector, this requires the frequency-dependent coefficient to vary negligibly across the carrier interval.

2.6 Comparison with direct tDCS and tACS

The steady TI source can be compared with transcranial direct-current stimulation (tDCS), and the beat source with tACS at the beat frequency. Both comparisons must specify the effective polarization entering the same sigmoid, rather than electrode current alone. For two equal TI carrier polarizations $\varepsilon_1 = \varepsilon_2 = \varepsilon > 0$, Eq. (5) gives the leading signed rate sources

$$Y_{\mathrm{DC}}^{\mathrm{TI}} \simeq Y_{\Omega}^{\mathrm{TI}} \simeq \frac{1}{2}\sigma''(v^*)\varepsilon^2.$$

A constant polarization ε_{D} and a sinusoidal polarization $\varepsilon_{\mathrm{A}} \cos \Omega t$, representing direct tDCS and tACS inputs at the same entry point, instead give

$$Y_{\mathrm{DC}}^{\mathrm{tDCS}} \simeq \sigma'(v^*)\varepsilon_{\mathrm{D}}, \qquad Y_{\Omega}^{\mathrm{tACS}} \simeq \sigma'(v^*)\varepsilon_{\mathrm{A}}.$$

Here ε_{D} is signed and a negative ε_{A} denotes a phase reversal. Matching either direct source to its TI counterpart therefore requires the equivalent polarization

$$\varepsilon_{\mathrm{eq}} = \frac{\sigma''(v^*)}{2\sigma'(v^*)}\,\varepsilon^2, \qquad \frac{\sigma''(v^*)}{\sigma'(v^*)} = -\rho \tanh\left[\frac{\rho(v^* - v_0)}{2}\right].$$

The ratio depends on sigmoid steepness and operating voltage; the rate ceiling $2e_0$ cancels. At the inflection the quadratic TI sources vanish.

For the JR parameters $\rho = 0.56$ mV^{-1}, $v_0 = 6$ mV, $e_0 = 2.5$ s^{-1}, and the pinned operating point $v^* = 8.30$ mV, $\sigma' = 0.4745$ s^{-1}mV^{-1} and $\sigma'' = -0.1508$ s^{-1}mV^{-2}. With each direct-input amplitude equal to the per-carrier TI amplitude, the source-magnitude ratio is $|\varepsilon_{\mathrm{eq}}|/\varepsilon = 0.1589\,\varepsilon/(1\text{ mV})$: 1.59% at 0.1 mV and 4.77% at 0.3 mV, rounded to 2–5%. At 0.3 mV per carrier, the equivalent direct polarization is -0.0143 mV, or -14.3 μV. The negative sign specifies a suppressive local source for tDCS and an inverted source phase for tACS.

These percentages match each TI carrier to the direct input. Matching the total TI peak instead requires a direct amplitude 2ε and halves the ratios. Matching RMS polarization requires tACS amplitude $\sqrt{2}\varepsilon$ or tDCS magnitude ε, since the two-tone TI RMS is ε. Direct tACS also generates a quadratic steady source, $\sigma''(v^*)\varepsilon_{\mathrm{A}}^2/4$; at equal RMS polarization it equals the direct TI steady source for this static detector. Equal electrode currents need not give equal polarizations because access differs between DC, the beat frequency, and the carriers, and between the compartments through which each acts.

Matching sources at the same entry point matches their leading circuit responses: the steady sources share the zero-frequency transfer, while the beat and tACS sources share the transfer at Ω. These two transfers need not be equal. The comparison is local and assumes a common operating state; the TI beat can also rectify after amplification and add to the mean shift (§4.6). It does not establish a universal ratio of physiological effects or equal spatial response maps.

2.7 Amplification: the synaptic low-pass and the network resonance

Synaptic filtering suppresses the carrier after the nonlinearity has generated the beat. Each JR excitatory synapse has transfer function $H_s(\omega) = Aa/(i\omega + a)^2$, a second-order low-pass with characteristic frequency $a/(2\pi) \approx 16$ Hz. It passes a 10 Hz component with relative amplitude about 0.7, while reducing a 100 Hz component to about 2.5%. Provided the carrier does not itself excite a responsive mode, the loop therefore reads predominantly the generated low-frequency drive.

What turns this recovered beat component into a large, frequency-selective response is the network. Linearizing the JR column (node) about its fixed point gives a leading eigenvalue pair $\mu_{\pm} = -\gamma \pm i\omega_0$; near the Hopf the response is dominated by this critical mode, with damping γ and

modal frequency ω_0. Below a supercritical Hopf, $\gamma \to 0^+$ and the column is a sharply tuned (high quality-factor $\mathcal{Q} = \omega_0/2\gamma$) resonator; driving it with the demodulated beat component (6) at Ω gives a resonant response. Writing G for the complex susceptibility, a constant-numerator two-pole approximation gives its magnitude as

$$|G(\Omega)| \propto \left[(\omega_0^2 + \gamma^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad |G(\omega_0)| \propto \frac{1}{2\gamma\omega_0} \quad (\gamma \ll \omega_0), \tag{11}$$

so the response peaks near ω_0 with an approximately Lorentzian power profile and $1/\gamma$ growth near a simple Hopf with nonzero input–output projections.[1] This weak-drive law requires the rectified mean shift to leave the state stable; above the Hopf, entrainment means capture of an autonomous rhythm and requires tracking its phase relative to the drive.

Write D for the detector's beat-source coefficient: for the JR rate source, $D = \frac{1}{2}\sigma''(v^*)$, half the local sigmoid curvature. Combining access, detection, and amplification gives

$$R_\Omega \simeq G(\Omega;\gamma)\; D\; \lambda_f(\omega_1)E_1\, \lambda_f(\omega_2)E_2, \tag{12}$$

where λ_f maps field to local carrier polarization, D converts the product of those polarizations into a beat drive, and G maps that drive to the complex observed response, whose measured amplitude is $A_\Omega = |R_\Omega|$. For JR, G is the response to a generated rate drive. A static detector has carrier-independent D at fixed polarization; a dynamic detector can retain frequency dependence. A carrier sweep and a beat sweep therefore constrain different stages rather than a single effective gain.

Resonance requires a beat source. The cubic Hopf normal form responds to an in-band drive but has no second-order difference-frequency source. A large resonant susceptibility alone therefore does not establish beat generation.

2.8 Carrier access depends on the effective membrane response

Carrier independence at fixed polarization does not imply independence at fixed applied field. As an isopotential access model, consider $H_m(\omega) = 1/(1 + i\omega\tau_m)$, with effective time constant τ_m. This is the response of a point membrane to injected current; a field coupled through a cable need not have the same transfer. For nearby carriers, the two polarizations acquire nearly the same attenuation. The quadratic beat then satisfies

$$\frac{|Y_\Omega(f_c)|}{|Y_\Omega^{\text{direct}}|} \simeq |H_m(\omega_c)|^2 = \frac{1}{1 + (\omega_c\tau_m)^2} \sim \frac{1}{(\omega_c\tau_m)^2} \quad (\omega_c\tau_m \gg 1), \tag{13}$$

where Y_Ω^{direct} is the beat generated with the same static coupling and no access attenuation. This filter belongs either in the field-to-polarization map or in the membrane equation that computes that polarization.

For the illustrative somatic time $\tau_m = 16$ ms, $|H_m|^2$ is about 10^{-4} at 1 kHz, 2.5×10^{-5} at 2 kHz, and 10^{-6} at 10 kHz. The assumed 0.2 ms sensitivity case instead gives 0.39 at 1 kHz and 0.14 at 2 kHz (Fig. S5); it has not been measured under TI at an AIS or node. Table 2 distinguishes measured responses, assumed access times, and downstream filtering. A measured axonal decay is not itself a field-to-polarization transfer or a TI cutoff.

A QIF reduction computes access within its membrane equation, so no additional membrane filter is applied to its current input. Short access times improve carrier transmission without specifying the mixing coefficient or circuit gain.

[1] Here ω_0 is the eigenvalue's imaginary part. The canonical oscillator has undamped frequency $\sqrt{\omega_0^2 + \gamma^2}$ and peak $\sqrt{\omega_0^2 - \gamma^2}$ for $\omega_0 > \gamma$; frequency-dependent network projections can shift the peak.

Response	Time scale	Evidence and role
Somatic point-membrane model	$\tau_m = 16$ ms; $f_{\text{char}} \approx 10$ Hz	Illustrative current-driven filter, informed by somatic measurements near 12 and 16.5 ms [4, 5].
Myelinated axon	minimum fitted decay ≈ 1 ms	Injected-current responses in Cohen et al.'s periaxonal circuit [7]; evidence for distinct axonal voltage dynamics.
Fast axonal access	assumed $\tau_m = 0.2$ ms; $f_{\text{char}} \approx 800$ Hz	Reduced-model sensitivity case for a fast compartment, such as an AIS or node; TI coupling and charging time require calibration.
Prior TI network model	$\tau_m = 1$ and 0.4 ms	Short effective membrane times governing kHz sensitivity in a gamma network [6]; model parameters, not measured axonal TI decays.
Field-induced charging	≈ 0.7 μs	Measured with strong microsecond field pulses in isolated neurons [23]; a different stimulation regime.
Cellular detector	channel- and state-dependent	Luff et al.'s TTX-sensitive mixing persists near 5 kHz [10]; this measures detection, not a passive time constant.
JR excitatory synapse	$\tau_s = 10$ ms; $f_{\text{char}} \approx 16$ Hz	Model filter acting on the generated rate, downstream of detection.

Table 2: Access, detection, and downstream filtering involve different response times. Measured decays, pulse-charging times, and assumed model parameters are identified separately. The characteristic frequency $f_{\text{char}} = 1/(2\pi\tau)$ is given only for the specified filters; an axonal decay is not automatically a TI cutoff. AIS, axon initial segment; TTX, tetrodotoxin.

A sufficiently long passive cable with a sealed end in a uniform axial field has quadratic-beat factors near 10^{-2} at 1 kHz and 10^{-3} at 10 kHz for $\tau_m = 16$ ms, exceeding the point-membrane values.

Channel kinetics contribute to the effective dynamic mixing response. Any separate gate filter needs a model or measurement and must represent kinetics absent from the detector. QIF, for example, lacks explicit sodium-channel kinetics. Neither QIF nor a static sigmoid specifies a universal 1 kHz cutoff. Carrier sweeps can constrain the combined access and mixing response, while beat-frequency sweeps probe the downstream amplifier.

2.9 Operating-point dependence

Because $Y_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far v^* sits from the sigmoid inflection. The Population Excitation–Inhibition Index (PEIX) of [24] measures signed departure from maximal sigmoid slope:

$$\text{PEIX}(v^*) = \text{sign}(v^* - v_0)\left\{1 - \text{sech}^2\left[\tfrac{\rho}{2}\,(v^* - v_0)\right]\right\}. \tag{14}$$

It is zero at v_0, positive above it, negative below it, and tends to ± 1 in the saturated tails. The normalized slope is instead $\sigma'(v^*)/\sigma'(v_0) = 1 - |\text{PEIX}(v^*)|$, with $\sigma'(v_0) = e_0\rho/2$. Demodulation follows the curvature, which vanishes at the inflection and in the tails and peaks in magnitude on the flanks. For the logistic curve, $\sigma'' = -\rho\sigma'\,\text{sign}(\text{PEIX})\sqrt{|\text{PEIX}|}$; the index specifies the side and saturation, but is not itself the mixing gain.

PEIX is a measurable operating-point index; its relation to mixing is specific to the memoryless logistic detector. In the Jansen–Rit detector, anything shifting v^* (for example, neuromodulation, attention, arousal, psychedelic exposure, or PV-interneuron dysfunction in Alzheimer's disease) scales, and can flip the sign of, the TI response; a dynamic detector responds through its own state-dependent coefficient and susceptibility, without a universal sign rule. State changes therefore test detection only when changes in recurrent gain are measured as well.

2.10 The exact mean field extends the static detector

The static rate curve applies when population activity follows the input fast enough. An exact mean field of quadratic integrate-and-fire (QIF) neurons makes this condition explicit and supplies the response outside that limit.

The corresponding population nonlinearity can be derived for all-to-all coupled QIF neurons with Lorentzian-distributed excitabilities. Their exact Montbrió–Pazó–Roxin (MPR) mean field evolves the firing rate r and mean voltage v [31]; adding dynamic synapses gives a second-generation neural mass model (NMM2) [32, 33]. With membrane time τ_m, physical firing rate r, and dimensionless voltage v,

$$\tau_m \dot{r} = \frac{\Delta}{\pi \tau_m} + 2rv, \qquad \tau_m \dot{v} = v^2 + \bar{\eta} - (\pi \tau_m r)^2 + I(t), \tag{15}$$

where $\bar{\eta}$ is the mean excitability, $\Delta > 0$ its half-width, and $I(t)$ the dimensionless input, including recurrent drive when present. Dots denote physical-time derivatives. For a constant total drive $Q = \bar{\eta} + I$, setting both derivatives to zero gives $v = -\Delta/(2\pi\tau_m r)$ and

$$\tau_m r = F(Q) = \frac{1}{\pi}\sqrt{\frac{Q + \sqrt{Q^2 + \Delta^2}}{2}}. \tag{16}$$

Primes on F denote derivatives with respect to the total drive Q. Below its inflection, curvature reflects recruitment of cells across threshold; above it, curvature reflects the concave square-root rate of active QIF neurons. This nonlinear curve exists without recurrent coupling. It has an inflection at $Q = \Delta/\sqrt{3}$ and an unbounded concave rise, unlike the saturating JR sigmoid.

If the population relaxes to this equilibrium faster than synaptic inputs change, NMM2 reduces to a static rate map with dynamic synapses [32, 33]. The rate then satisfies $r(t) \simeq F(Q(t))/\tau_m$, with recurrent drive filtered by the second-order synaptic operator of Eq. (B5) in Appendix B. Here τ_s is the synaptic time constant. The population relaxation time τ_{rel} must satisfy $\tau_{\mathrm{rel}} \ll \tau_s$. Applying the same static map directly to a carrier of angular frequency ω_c requires the stronger condition $\omega_c \tau_{\mathrm{rel}} \ll 1$. A population can therefore be quasistatic for its synaptic inputs yet dynamic for a kHz carrier.

Recurrent excitatory–inhibitory circuits can resonate without intrinsic cellular resonance [13, 14]. In the fast-membrane limit, synaptic modes remain, whereas resonances requiring finite membrane dynamics need not survive.

Outside the adiabatic limit, the quadratic voltage term mixes the carriers before the rate responds. For an uncoupled population under a weak current $I_{\mathrm{ac}} \cos \omega t$, write the mean-rate shift as $I_{\mathrm{ac}}^2 \kappa_{\mathrm{DC}}/\tau_m$. At slow carriers, $\kappa_{\mathrm{DC}} \to F''(Q)/4$; at fast carriers, $\kappa_{\mathrm{DC}} \sim F'(Q)/[2(\omega\tau_m)^2]$. The slow response follows curvature and can be negative above the inflection. The fast DC coefficient instead follows the positive slope F', largest at the inflection, without a PEIX-based sign reversal: voltage excursions of order $I_{\mathrm{ac}}/(\omega\tau_m)$ generate a positive input shift through v^2. A shorter membrane time can therefore change both the magnitude and sign of the mean response at a given physical carrier frequency.

The fast-carrier attenuation already includes membrane access. Dividing by the squared voltage excursion leaves a coefficient tending to $F'/2$. For two nearby tones, the voltage quadratic generates a beat source proportional to $I_1 I_2/(\omega_1 \omega_2 \tau_m^2)$, where I_j is the current amplitude of tone j, with the same product law as the static detector. The dynamic rate equation attenuates the carriers once more, after this mixing; it does not add that attenuation to the generated beat. The crossover coefficient is derived in Appendix B.2; fixed-voltage coefficients and carrier exponents are given in the supplement (§S2.1).

For the QIF mixer, the role of $\sigma''/2$ is played by the unit coefficient of v^2 in membrane-time units. The voltage and rate dynamics that convert this source into a rate beat belong to the

response G. QIF thus provides a cellular realization of quadratic detection, with its own frequency dependence. Other smooth cellular mechanisms can contribute to the same effective second-order coefficient. Their weak-input product law does not fix the coefficient's magnitude, phase, or carrier dependence.

3 Methods

A *forced response* is the beat-driven component measured by lock-in; a purely forced regime requires stability under stimulation. *Entrainment* means frequency and phase capture of a self-sustained oscillator [34]. Our frequency-dominance screen does not establish that capture. *Gating* means modulation of a persisting rhythm's amplitude [35, 36]; a rate fold alone does not establish it. Detailed criteria are in the supplement (§S7).

3.1 Stimulation waveforms and matched-source controls

Dense perturbative sweeps use the three-line waveform

$$s(t) = \varepsilon\,[\,1 + m\cos\Omega t\,]\cos\omega_c t, \qquad \omega_c = 2\pi f_c, \tag{17}$$

with dimensionless modulation depth m. This AM surrogate parameterizes dense resonance and coupling sweeps by one amplitude. It contains a central carrier line absent from genuine TI, but matches its leading beat source: $Y_\Omega = \frac{1}{2}\sigma''\varepsilon^2 m$. For matched genuine tones, $\mathrm{LPF}[s^2] = \varepsilon^2(1+\cos\Omega t)$; for the $m = 1$ surrogate, $\mathrm{LPF}[s^2] = \varepsilon^2(\frac{3}{4} + \cos\Omega t + \frac{1}{4}\cos 2\Omega t)$. Thus the mean source differs, and the surrogate adds a second harmonic at 25% of the fundamental. We use genuine tones for the demodulation, curvature, square-law, laminar, and kHz demonstrations, and wherever mean shifts or harmonics enter the comparison: timing versus rate and the TI/tACS and NMM2 endpoint controls. Open-loop agreement at the beat is checked at one operating point; it does not imply equal driven operating states.

3.2 Amplitude scaling and harmonic measurements

We sweep amplitude at fixed coupling and operating state, measuring the fundamental and second harmonic to test the laws of §2.4.

The response at the envelope frequency is read out with a software lock-in amplifier: multiplying $v(t)$ by sine and cosine at Ω and averaging estimates the component at that frequency. A finite window suppresses neighboring frequencies without eliminating them. With a Hann window, the in-phase ($L_{\cos}$) and quadrature ($L_{\sin}$) components are

$$L_{\cos} = \frac{2}{\sum_k w_k}\sum_k w_k\,(v_k - \bar v)\cos\Omega t_k, \quad L_{\sin} = \frac{2}{\sum_k w_k}\sum_k w_k\,(v_k - \bar v)\sin\Omega t_k,$$
$$A_\Omega = \sqrt{L_{\cos}^2 + L_{\sin}^2}, \tag{18}$$

with sample index k, sample times t_k, voltages $v_k = v(t_k)$, and Hann weights w_k. The windowed mean is $\bar v = \sum_k w_k v_k / \sum_k w_k$. For a test tone $v = A_{\mathrm{test}}\cos(\Omega t + \theta_0)$, with amplitude A_{test} and phase θ_0, the long-window limits $(L_{\cos}, L_{\sin}) \to A_{\mathrm{test}}(\cos\theta_0, -\sin\theta_0)$ give $A_\Omega \to A_{\mathrm{test}}$: the normalization returns the physical amplitude, twice the magnitude of the windowed Fourier coefficient. Subtracting the windowed mean suppresses leakage from the much larger baseline into the small beat estimate (supplement, §S7.1).

Measure both harmonics over the same settled window. Correct their amplitude ratio by the gain ratio $|G(2\Omega)/G(\Omega)|$. The amplitude sweep tests linear versus quadratic scaling. Estimating a mixture coefficient through Eq. (S3) assumes the mixed detector family and its sign branch; a threshold detector can match one harmonic ratio.

3.3 Head model and field computation

We compute maps in IXI002, an open-access T1-weighted MRI of a 35-year-old woman, using the hippocampal montage of Violante et al. [37]. The six-tissue SimNIBS conductor is quasi-static with no kHz dispersion correction [38]. Pair fields are solved once and scaled to peak currents summing to 4 mA: $2 + 2$ mA for 1:1 and $1 + 3$ mA for 1:3. The posterior share is swept from 0.1 to 0.9. Conductivities, electrodes, solver settings, and sensitivity protocols are in the supplement (§S3.3).

Local polarization is $\varepsilon_j = \lambda_0 I_j^{\text{pair}} \mathbf{F}_j \cdot \mathbf{n}$ with uniform $\lambda_0 = 0.5$ mV per V/m [39] and field per unit current $\mathbf{F}_j$. This coupling sets absolute amplitudes but cancels from normalized maps, focus positions, and target-to-reference ratios. It is an assumed field-to-polarization conversion, not a measured AIS transfer.

At each node, $\mathbf{n}$ maximizes the envelope at 1:1 and is then held fixed for every ratio and every map. This geometric direction need not match a cell's orientation. Radial and isotropically averaged orientations provide sensitivity comparisons; the latter averages the signed beat before taking its magnitude, allowing cancellation across orientations.

The target is the left hippocampus segmented with FreeSurfer (1,568 gray- and white-matter nodes). The reference comprises 273,544 gray-matter nodes outside that ROI, including other deep gray matter. The target-to-reference contrast is the ROI 95th percentile divided by the reference 95th percentile; the mean-based companion replaces the numerator by the ROI mean. Nodes receive equal weight. Square roots of second-order maps provide degree-matched geometric contrasts; the raw maps remain the detector predictions before gain.

The focus is the centroid of the top 10% of ROI nodes ranked by each map. Displacement is measured along the hippocampal long axis, with anterior positive. Sensitivity checks vary montage, atlas label, gray/white-matter inclusion, focus fraction, and orientation. These geometric maps precede detection and recurrence: $Y_{\text{DC}} = \sigma''(v^*)\mathcal{I}/2$ is a signed rescaling only for uniform curvature. The head model supplies neither operating states nor a calibrated AIS transfer.

3.4 Single-column Jansen–Rit: integration, lock-in, and controls

We integrate the JR column as six first-order ordinary differential equations. The TI field enters only the pyramidal output sigmoid, Eq. (B2), as $s(t)$, while the readout $v = y_1 - y_2$ contains only synaptic potentials. A beat in this readout must therefore have passed through the nonlinearity and the recurrent loop. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

Integration uses fixed-step fourth-order Runge–Kutta (RK4), with time steps chosen for each carrier band. Halving each step changes the reported lock-in amplitudes by less than 1%. The resonance sweep settles for 10 s and measures for 20 s; run-specific steps and windows are listed in the supplement (§S7.2). The quoted 10 kHz attenuation is analytic, not simulated.

Numerical continuation gives the fixed-point and limit-cycle branches in Fig. 3. Independently solving the resting fixed point and tracking the leading pair of its 6×6 Jacobian locates the upper Hopf at $p_H \approx 315.70$ s^{-1} and $f_0 \approx 11.16$ Hz, consistent with [40]. For $p > p_H$ the fixed point is stable; the 11.1 Hz example drives it near resonance.

The open-loop sigmoid at fixed v^* isolates detection; its signed lock-in tests the curvature law. The closed-loop column measures recurrent gain over an (Ω, p) grid. Controls test quadratic amplitude scaling on a well-damped focus, abolition of the beat when the pyramidal sigmoid is replaced by its tangent, and agreement of σ' and σ'' with finite differences. Field-free trajectories initialized near the fixed point check the continuation.

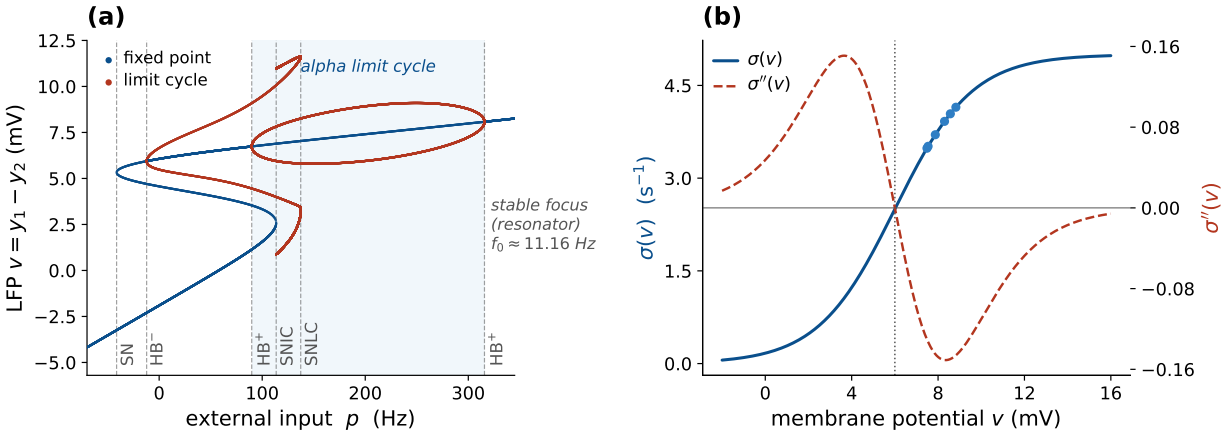


Figure 3: JR operating states and sigmoid curvature. *(a)* Numerical continuation of fixed points (blue) and cycle extrema (red) against external drive p; solid branches are stable, faded branches unstable. The alpha cycle lies between Hopf points near 90 and 316 s^{-1}. Beyond the upper Hopf the fixed point is a stable focus, with $f_0 \approx 11.16$ Hz. Markers identify saddle-nodes (SN), Hopf points (HB$^{\pm}$), the saddle-node on an invariant circle (SNIC), and the saddle-node of limit cycles (SNLC). *(b)* Sigmoid and curvature; dots mark the modeled operating points.

3.5 Exact next-generation mean field (NMM2 PING)

The pyramidal–interneuron network gamma (PING) circuit couples excitatory and inhibitory MPR populations through second-order synapses [31, 32, 33]. We use the Rosetta Stone formulation [27], which places this derived mean field on the same footing as the JR and laminar models. For population $X \in \{E, I\}$ (excitatory or inhibitory), r_X is the physical firing rate (ms^{-1} in the integration), v_X the dimensionless mean QIF voltage, and τ_X the membrane time (ms). Dots denote physical-time derivatives.

$$\begin{aligned} \tau_E \dot{r}_E &= \Delta/(\pi\tau_E) + 2 r_E v_E, & \tau_E \dot{v}_E &= v_E^2 + \bar{\eta} - (\pi\tau_E r_E)^2 + \tau_E J\,(r_E^{\mathrm{syn}} - r_I^{\mathrm{syn}}) + I_E(t), \\ \tau_I \dot{r}_I &= \Delta/(\pi\tau_I) + 2 r_I v_I, & \tau_I \dot{v}_I &= v_I^2 + \bar{\eta} - (\pi\tau_I r_I)^2 + \tau_I J\,(r_E^{\mathrm{syn}} - 2 r_I^{\mathrm{syn}}), \end{aligned} \tag{19}$$

with second-order synaptic activations $r_E^{\mathrm{syn}} = \hat{K}_a[r_E]$ (excitatory, time constant τ_a) and $r_I^{\mathrm{syn}} = \hat{K}_g[r_I]$ (inhibitory, τ_g), each obeying the same critically damped operator as the JR synapse, $\tau_s^2 \ddot{r}_{\mathrm{syn}} + 2\tau_s \dot{r}_{\mathrm{syn}} + r_{\mathrm{syn}} = r$; here J is the dimensionless global synaptic gain and the relative weights $(w_{EE}, w_{EI}, w_{IE}, w_{II}) = (1, 1, 1, 2)$ use target X and source Y in w_{XY}. The only weight asymmetry is the doubled inhibitory self-coupling, $w_{II} = 2$.

In the schematic, $\mathcal{U}_j(t)$ is the total synaptic and field input to QIF neuron j, and J^* denotes a coupling-induced Hopf. The common background drive $\bar{\eta}$ is the bifurcation parameter: raising it past a supercritical Hopf at $\bar{\eta}_H = 1.110$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory voltage equation as a current-like forcing, $I_E(t) = \lambda_I E(t)$ in the convention of the exact mean field [32], with instantaneous field-to-current coupling λ_I, as two genuine tones at $f_c \mp \Delta f/2$ (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) of current amplitude A_f in the dimensionless MPR drive units, the same normalization as $\bar{\eta}$. Thus λ_I is measured in normalized drive per V/m. The E membrane $\tau_E \dot{v}_E$ integrates this forcing into the carrier-frequency excursion $A_f/(\omega\tau_E)$ that the exact quadratic v_E^2 term rectifies, with no postulated sigmoid; the PING as run therefore computes membrane access with $\tau_m = \tau_E$.

We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (18)) of r_E at Δf over the $(\Delta f, \bar{\eta})$ plane. Parameters (Rosetta Stone PING [27]): $\Delta = 1$, global coupling $J = 15$, $\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar{\eta}$ swept through the supercritical Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz), located by bisection on the leading eigenvalue of the mean-field Jacobian. The

simulations retain these finite membrane times; Eq. (B5) characterizes the fast-membrane limit.

As in JR, either background drive or coupling can bring the circuit to its Hopf. The resonance map sweeps $\bar{\eta}$ (Fig. S8); the coupling sweep, called the J-curve, varies J at fixed $\bar{\eta}$ toward J^*, located by bisection on the leading Jacobian eigenvalue, and reads the lock-in peak of r_E over Δf (Fig. S10). Unlike the JR J-curve, this sweep does not pin the operating point. At $\bar{\eta} = 0$, we track the eigenpair that reaches the Hopf by continuation and define γ from that critical pair, even where another mode dominates the response. The linear reference includes the state-dependent second-order source and full recurrent transfer. Search grids, amplitude checks, and normalizations are in the supplement (§S4.2).

Stability of the driven endpoint is checked using period-to-period perturbation multipliers, direct perturbation decay, and a finite-time Lyapunov estimate; settings and results are given in the supplement (§S4.3).

3.6 QIF spiking network (microscale of NMM2)

NMM2 is the exact infinite-population mean field of all-to-all QIF neurons [31, 27, 33]. We integrate that network directly to compare beat-locked and ongoing activity at the spike level: $N = 8000$ QIF neurons per population, parameters identical to the PING mean field (Eq. (19)), with single-neuron voltages V_j, finite spike threshold $V_{\text{peak}} = 100$ and reset $-V_{\text{peak}}$ in model voltage units, excitabilities $\eta_j \sim \text{Lorentzian}(\bar{\eta}, \Delta)$ sampled by deterministic quantiles, second-order synapses driven by the empirical population rates, and a genuine two-tone field entering the excitatory membranes. Spiking and threshold dynamics are where the AM surrogate and the physical waveform cannot be assumed equivalent beyond perturbation theory, so this run uses two tones throughout.

The common background drive $\bar{\eta}$ places the field-free mean field below or above a gamma Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz). We use $\bar{\eta} = 0.85$ with a 55 Hz beat and $\bar{\eta} = 1.5$ with a 42 Hz beat. Both runs have carrier $f_c = 300$ Hz and amplitude $A_f = 35$ per tone. We label the conditions by their field-free state and separate the beat-locked component from other gamma activity. Spectra and rate folds resolve frequency components but do not establish phase capture.

We also run both conditions at $A_f = 10$ per tone with the same measurement settings. For $\bar{\eta} = 0.85$, separate checks double the population size, double the measurement window, or halve the integration step; further runs raise V_{peak} to 200 and 400 at both time steps. An equal-RMS mean-field control replaces the two tones by one 300 Hz carrier with peak amplitude $\sqrt{2}A_f$, using the same duration: both inputs have mean-square amplitude A_f^2 over these windows. The carrier comparison starts from the field-free equilibrium and uses 800 ms settling and 800 ms measurement, with a 3200 ms window to check spectral leakage at the larger amplitude.

The carrier-averaged stability calculation includes the drive-dependent second-order DC source (supplement, §S4.6). Direct two-tone integrations retain the beat and separately check the averaged prediction.

The network is integrated with Euler steps of 0.0025 ms, settled for 800 ms (over four field-free relaxation times at $\bar{\eta} = 0.85$, where the mean-field relaxation time is 176 ms) and measured over an integer number of beat cycles, 44 at 55 Hz and 34 at 42 Hz (800 and 810 ms), removing the phase-coverage bias from incomplete beat cycles. Envelope-locked timing is quantified by the quadrature lock-in A_Ω of $r_E(t)$ at Δf, reported as the modulation depth $A_\Omega/\langle r_E \rangle$ (a stable measure; the off→on lock-in ratio would divide by the field-free floor), and per neuron by the pairwise phase consistency of each neuron's spike phases at Δf (PPC; [41]), which is unbiased in the spike count, over neurons with at least ten spikes in the window.

A cutoff-free check uses the equivalent theta-neuron equation, $V_j = \tan(\theta_j/2)$, with interpolated crossings of $\theta_j = \pi$. Separate step, population, and window checks test a predeclared 5% rate-

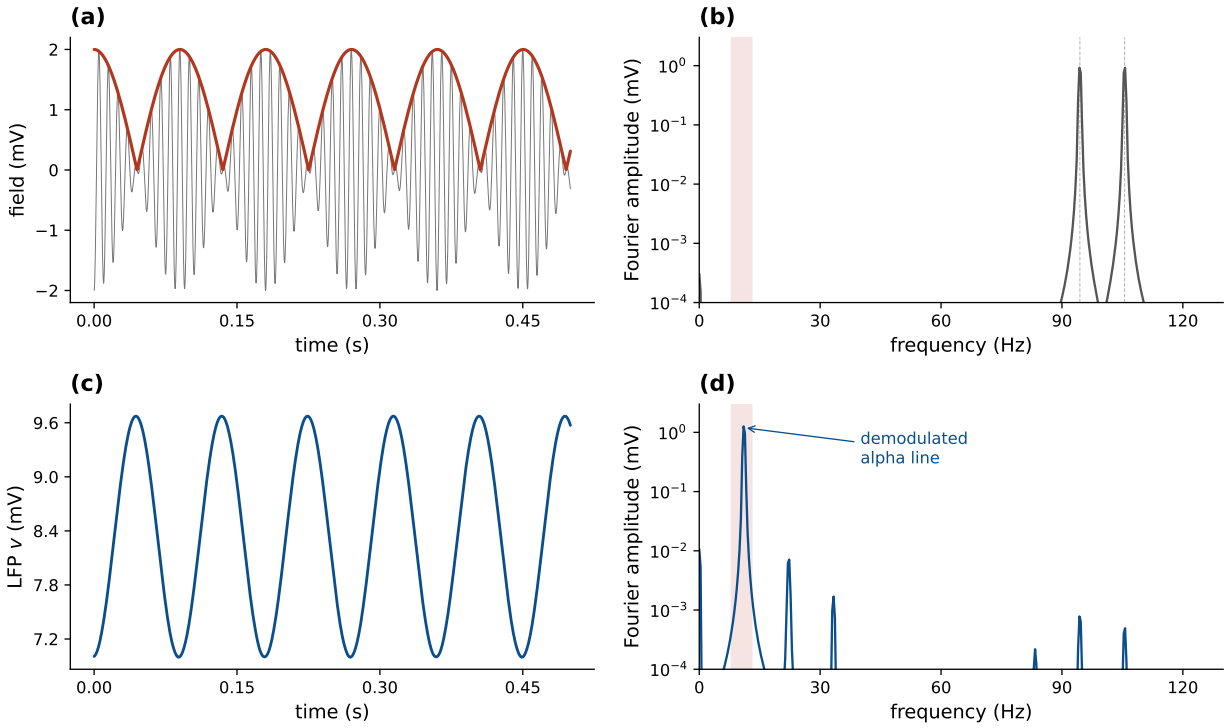


Figure 4: The JR column generates a beat absent from the input. *(a)* Two equal tones at 94.45 and 105.55 Hz and their envelope, $2\varepsilon|\cos(\Omega t/2)|$. *(b)* Input spectrum: only the two tones are present. The alpha-band floor is 1.6×10^{-8} of their amplitude, from finite-window leakage. *(c,d)* The synaptic voltage $v = y_1 - y_2$ and its spectrum show the generated 11.1 Hz beat. The field-free column is a stable focus near its alpha Hopf.

depth tolerance (supplement, §S4.5). Quantitative convergence remains unestablished; the finite network is interpreted qualitatively.

3.7 Coupling sweeps, timing, and the tACS control (Jansen–Rit)

The JR sweeps vary coupling while keeping detection fixed. For each connectivity C, we solve for the external drive $p(C)$ that holds the field-free voltage at $v^* = 8.30$ mV, then obtain damping and modal frequency from the Jacobian. This fixes the background curvature without canceling stimulation-induced mean shifts.

The gain sweep uses the AM surrogate; the timing comparison uses genuine two tones and measures modulation and mean of the same pyramidal rate $r_P = \sigma(v + s)$. The tACS control uses a direct sinusoid at the same beat frequency and reads the same voltage lock-in as TI. Further controls match tACS to TI's generated beat source and repeat the comparison with longer settling and measurement windows. Waveforms, amplitudes, and run windows are recorded together in the supplement.

4 Results

4.1 The JR column generates and resonantly amplifies the beat

An input containing only the two tones $f_c \mp \Delta f/2$ produces an output oscillation at their 11.1 Hz beat, near the column's alpha resonance (Fig. 3). The alpha peak is absent from the input spectrum (Fig. 4).

In the stable-focus regime ($p > p_H$), the response is approximately Lorentzian and its peak grows from 0.23 to 0.43 to 0.70 mV toward the Hopf (Fig. 5), as expected from the $1/\gamma$ susceptibility.

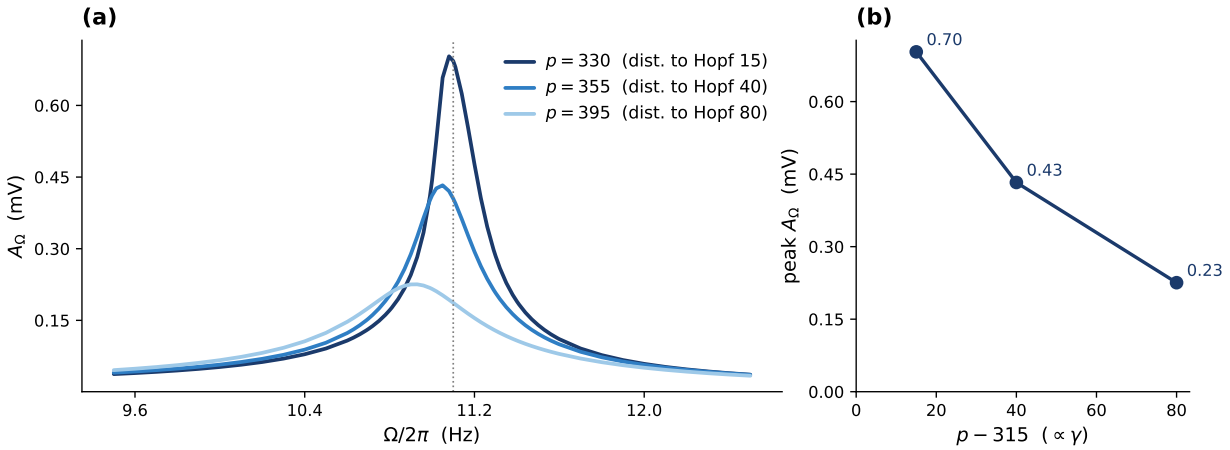


Figure 5: The JR beat resonance sharpens toward the Hopf. Stable-focus response at $\varepsilon = 0.3$ mV. *(a)* Lock-in amplitude against beat frequency; the finite-amplitude peak moves from 11.08 to 10.93 Hz across the curves. *(b)* Peak amplitude increases as p approaches the upper Hopf from above. These three points show sharpening and growth; the coupling sweep quantifies the $1/\gamma$ scaling and its saturation.

Without a drive the focus is silent at Ω; the nonlinearity generates that component and recurrence amplifies it. The peak shifts with the driven operating state; the near-Hopf law gives its leading gain, not its exact frequency. The coupling sweep tests that law over a decade of damping. On the alpha branch below p_H, the column free-runs at f_0, so the lock-in can contain both driven and autonomous activity. The dominant output frequency f_{out} then identifies a frequency-dominance region (Fig. S4), without establishing phase capture.

4.2 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band, for both the open-loop detector and the closed-loop column near the Hopf (Fig. 6). As a function of the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above the corner in the point-membrane model (Fig. S5). Its 16 ms response gives beat attenuation of about 10^{-4}–10^{-6} at 1–10 kHz; the cable-end case in the passive-cable calculation gives about 10^{-2}–10^{-3}. The assumed 0.2 ms point-membrane response retains appreciable carrier access.

We also drive the full recurrent column with a kHz field, passing it through an explicit membrane low-pass (an extra first-order state, time constant τ) for each modeled compartment before its polarization reaches the sigmoid. The column demodulates and resonates at the recovered alpha in the fast-access case ($\tau = 0.2$ ms), as Eq. (13) predicts (Fig. S9a). With two genuine tones centered at 2 kHz and beating at 10.914 Hz, the output contains the beat absent from the input (Fig. S6); the analytic access factor is $|H_m|^2 = 0.137$ for $\tau = 0.2$ ms. This quantifies the fast-access hypothesis under the specified response time and coupling.

Figure 7 converts this access assumption into a source budget. With $\varepsilon_j = \lambda_0 E_j |H_m(\omega_j)|$, the equivalent direct beat input is $\varepsilon_{\text{eq}} = Y_\Omega/\sigma'$, before recurrent gain. At 2 kHz, $\tau_m = 0.2$ ms, $\lambda_0 = 0.5$ mV per V/m, and 0.5 V/m per tone, it has magnitude 1.36 μV; the steady equivalent is nearly identical. Reducing the assumed coupling to 0.15 scales these sources by 0.09. Thus the matched-polarization 2–5% comparison does not carry over unchanged to matched applied fields.

At equal fields, the 0.2 ms point-membrane model predicts a narrowband 9/1 kHz source ratio of 0.020. The limit 1/81 applies only when both bands lie above their access corners. A sealed long cable instead approaches a 1/9 ratio. These calculations locate conditional input budgets; they do not calibrate the biological transfer. The dynamic QIF equations already include membrane access, so this filter must not be applied again.

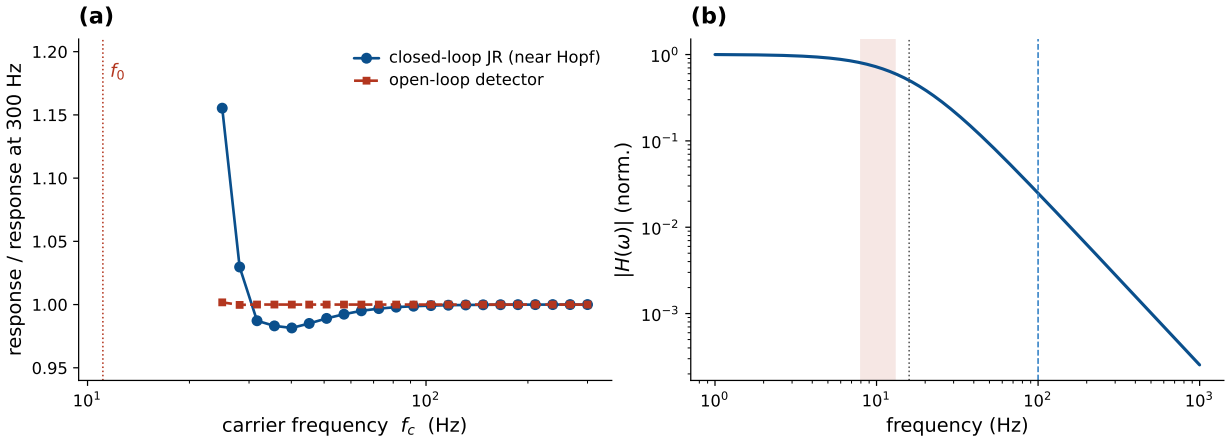


Figure 6: The demodulated response is carrier-independent at fixed polarization (Jansen–Rit). *(a)* The demodulated response is nearly flat across carrier frequency $f_c = 25$–300 Hz for the open-loop detector and for the closed-loop JR near the Hopf, with deviations as f_c approaches the natural band. Each curve is normalized by its own response at 300 Hz: 0.00670 s^{-1} open loop and 0.512 mV closed loop. Their ratio is a rate-to-voltage transfer of 76.4 mV s. The open-loop carrier independence follows to leading order from Eq. (6). *(b)* The second-order-synapse low-pass $|H_s(\omega)|/|H_s(0)|$ passes the envelope ($\sim$0.7) and suppresses the 100 Hz carrier ($\sim$2.5%). Dotted line: synaptic corner $a/2\pi \approx 16$ Hz; shaded band: envelope (8–13 Hz); dashed line: 100 Hz carrier.

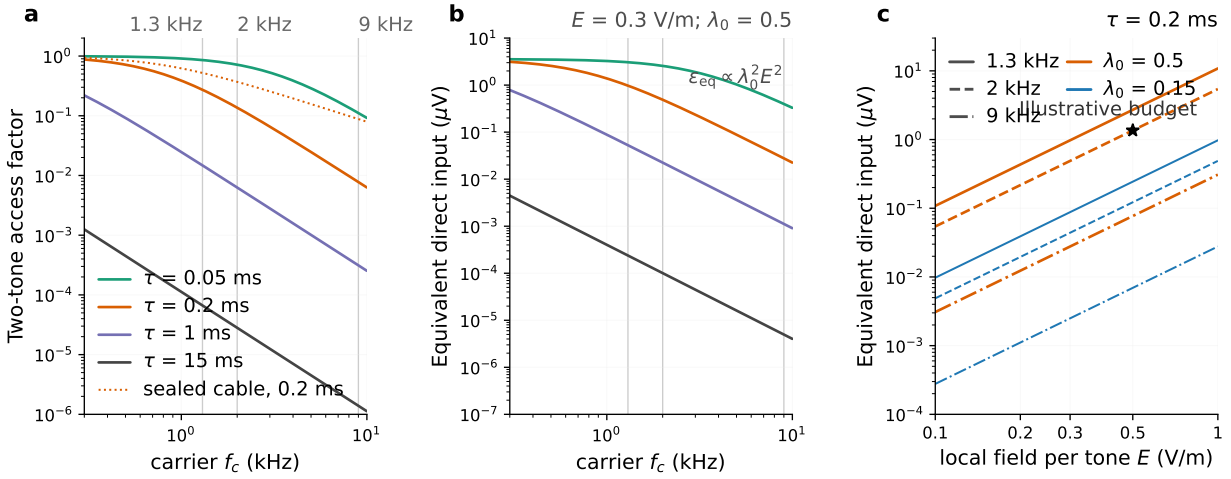


Figure 7: Carrier access sets the source budget at fixed field. *(a)* Exact two-tone access factor for tones 10 Hz apart; dotted, sealed-cable approximation at 0.2 ms. *(b)* Beat-equivalent direct polarization at the pinned JR state, $E = 0.3$ V/m per tone and $\lambda_0 = 0.5$ mV per V/m. For $\lambda_0 = 0.15$, multiply by 0.09. *(c)* Field dependence at 0.2 ms for both couplings; star marks the illustrative budget. All sources are leading-order, before recurrent gain; λ_0 and the access time are assumptions.

4.3 The nonlinearity is the detector: the σ'' curvature law and its controls

The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the operating range, vanishing and reversing sign at the inflection v_0 (Fig. 8)—demodulation is the sigmoid-curvature term of Eq. (6), and the closed-loop operating points sit where σ'' is large and negative. Changing its sign reverses the source phase; at matched network susceptibility, its magnitude sets the same resonant-gain calculation on either flank.

In the recurrent column, the response scales quadratically at small drive (log–log slope 1.99, measured as in §3.2), and linearizing the field-receiving sigmoid reduces it from 1.34 to 0.0021 mV (Fig. 8b,c). The open-loop surrogate comparison in panel (d) supports its use for leading-order beat sweeps, with the waveform and near-Hopf limitations specified in §3.1.

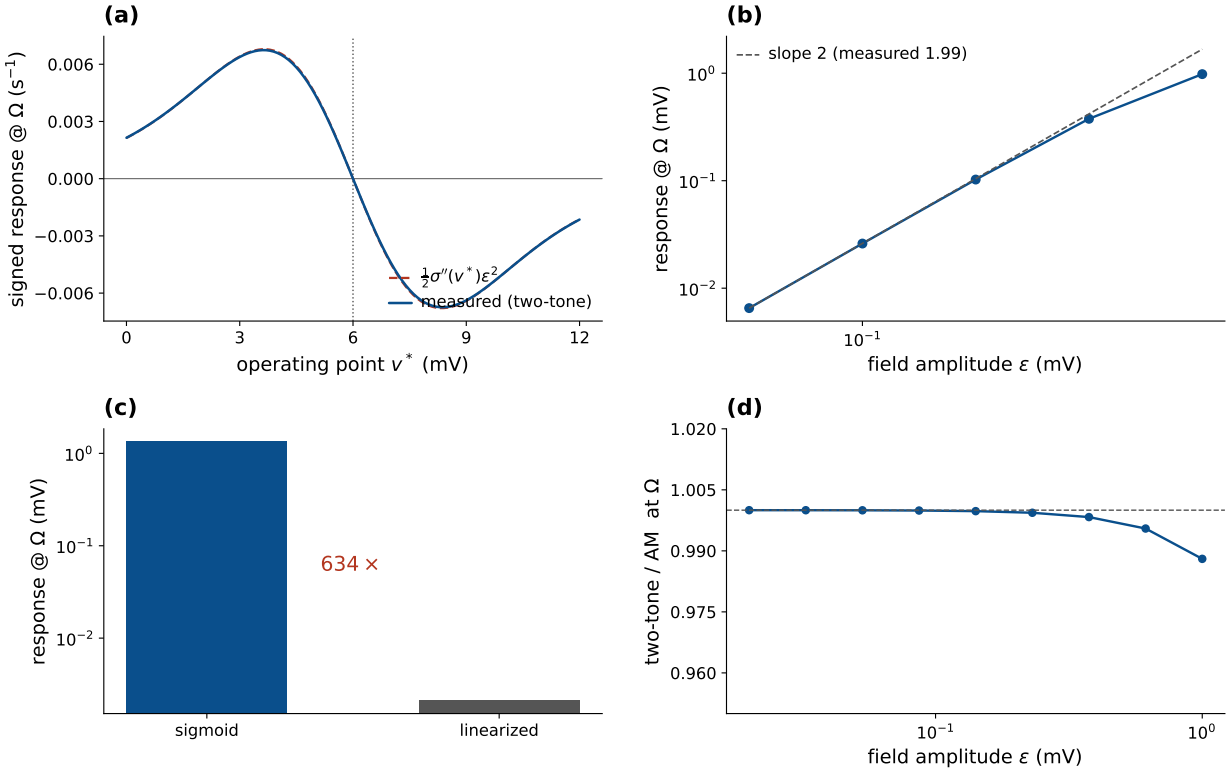


Figure 8: Curvature determines the JR detector. *(a)* The signed open-loop beat follows $\sigma''(v^*)\varepsilon^2/2$ within 0.9%, vanishing and reversing sign at v_0. *(b)* The recurrent response has amplitude slope 1.99. *(c)* Linearizing the receiving sigmoid reduces the beat 634-fold. *(d)* Open-loop two-tone/AM beat ratio at $v^* = 8$ mV: 1.0000 at 0.02 mV and 0.988 at 1 mV. This checks the leading local source, not the nonlinear recurrent response near the Hopf.

4.4 Realistic-head comparison of envelope and quadratic maps

In this single-subject geometric illustration, changing the current ratio moves the envelope and mean-square ROI centroids in opposite directions, while the quadratic beat map only rescales. Each centroid uses the top 10% of hippocampal nodes; it is not a whole-brain maximum. Figure 9 compares the maps of Table 1 for the Violante montage in IXI002, using the same polarizations and fixed 1:1 projection (§3.3).

Table 3 summarizes the 1:1 to 1:3 change. The pair-current product falls from 4 to 3 mA^2, so the quadratic excursion falls to 75% of its value at every node, to machine precision. Its normalized map, contrast, and focus are unchanged. The envelope focus moves anteriorly, toward the pair whose current falls; the mean-square focus moves posteriorly, toward the pair whose current rises. At 1:1, degree matching reduces the quadratic contrast to 1.08, below the envelope's 1.22, while mean-square polarization favors the reference. These modest contrasts separate geometric predictions but do not establish physiological deep selectivity.

Table 3: Hippocampal maps under current-ratio changes. Means refer to the ROI; contrasts are ROI/reference 95th-percentile ratios. Parentheses give square-root, degree-matched contrasts. Focus shifts use the top 10% of ROI nodes, with anterior positive; ranges over other fractions are supplementary.

Map	Mean, 1:1	Contrast, 1:1	Contrast, 1:3	Mean change	Focus shift
Envelope	0.16 mV	1.22	0.95	−45%	+2.5 mm
Quadratic beat	0.018 mV^2	1.17 (1.08)	1.17 (1.08)	−25%	0 mm
Mean square	0.0094 mV^2	0.92 (0.96)	0.90	+24%	−4.5 mm

The sweep shows two kinds of steering (Fig. 9e,f). The envelope ratio peaks near balance (1.24 at a

posterior share of 0.525) and falls to 0.89–0.94 at the extremes. Its focus moves only while the two carriers are comparable inside the hippocampus. At a posterior share of 0.6 or more, the anterior pair is the weaker carrier at more than 96% of hippocampal nodes; at 0.4 or less, the posterior pair is. The envelope map then equals twice the weaker pair's polarization over most of the ROI, and its measured focus changes little with further imbalance. The focus of $\mathcal{I}$ moves continuously instead, from 1.5 mm anterior at a posterior share of 0.1 to 6.3 mm posterior at 0.9, while its ratio stays between 0.79 and 0.94 (0.89–0.97 degree-matched). The quadratic excursion keeps one ratio and one focus across the whole sweep.

The isotropic beat map (Fig. 9d,g) has a target-to-reference ratio of 1.18 (1.08 degree-matched), close to that of the fixed-direction quadratic map. It also scales with $I_1^{\text{pair}} I_2^{\text{pair}}$, so the current ratio cannot steer it either.

Across all 32 combinations of montage, label, tissue definition, and focus fraction, the envelope shift remains anterior, the mean-square shift posterior, and the quadratic beat shift zero. Radial projections preserve those directions but change absolute contrasts more than the atlas label does (supplement, §S3.4).

4.5 Coupling sets the gain at fixed detection

Changing connectivity at a fixed field-free operating point isolates recurrent gain from detection. We vary C and adjust the external drive $p(C)$ to hold v^*, its curvature $\sigma''(v^*)$, and the open-loop beat amplitude $A_{\text{open}} \simeq \frac{1}{2}|\sigma''(v^*)|\varepsilon^2 m$ constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$) the damping $\gamma \to 0$, the demodulated response grows $\sim$30$\times$ and the envelope resonance sharpens (Fig. 10a,b), a network effect since the local detector curvature is fixed while the recurrent transfer changes. Its modal frequency and input–output projections vary with C as well as its damping. The rate-to-voltage gain A_Ω/A_{open} (mV s) tracks $1/\gamma$ before saturating near criticality (Fig. 10c); driving the same column to the Hopf through the input p rather than the coupling C gives a similar leading gain–γ trend. Both paths approach a weakly damped mode; their finite-distance gains need not coincide because the modal projections can differ.

4.6 The resonance amplifies the envelope-locked modulation more than the mean rate

Pinning v^* to hold the local detector curvature fixed, as in Fig. 10, and driving the column toward the Hopf through the coupling, we read both quantities from the same observable, the pyramidal rate $r_P = \sigma(v + s)$ with the field in its argument. With genuine two tones, its beat-locked modulation at $\Delta f = f_0$ grows 13.2$\times$ (26 to 339 mHz), tracking $1/\gamma$ before saturating; its mean shift grows 4.4$\times$ (-1.8 to -7.7 mHz), but stays below 0.2% of the 3.9 Hz baseline (Fig. S11). The ratio of modulation to mean-shift magnitude rises from 14.5 far from the Hopf to 58 near $\gamma = 1\ \text{s}^{-1}$, then falls to 44 closest to the Hopf as rectification of the amplified oscillation accelerates the mean shift. The surrogate comparison is given in the supplement (§S2.3). A small mean-rate change can therefore accompany a much larger beat response in this weakly driven regime, consistent with primate observations [20]; both effects are compared in the same rate units.

The demodulator feeds the network a constant term and a line at the beat Ω (Eq. (5)), and the transfer $G(\Omega)$ of Eq. (11) treats them differently: $|G(\omega_0)| \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while $|G(0)| \propto 1/(\omega_0^2 + \gamma^2)$ stays finite. This comparison describes the response to the detector's DC and beat sources. The growing beat response itself rectifies and contributes another mean shift, quadratic in its amplitude. Before saturation that contribution can grow as $1/\gamma^2$, so bounded $G(0)$ does not imply a bounded mean shift throughout the approach to the Hopf.

4.7 TI and tACS share the linear amplifier

TI and direct low-frequency stimulation share the same linear amplifier. Matching their generated in-band drives therefore matches their linear outputs; resonance alone gives TI no efficiency ad-

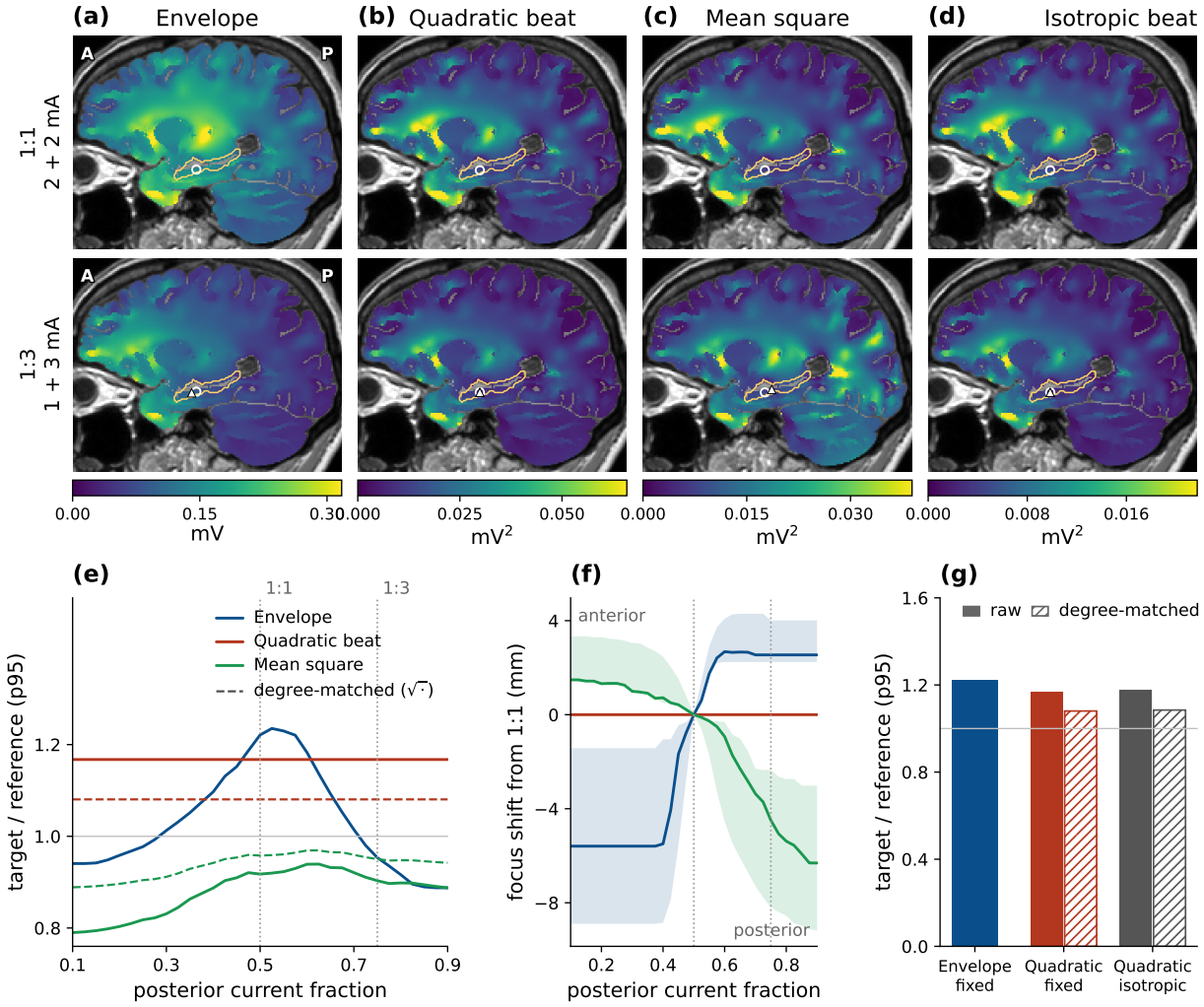


Figure 9: Envelope and quadratic maps in IXI002. Violante montage; left hippocampus outlined in gold, anterior to the left. *(a–d)* Sagittal maps at $2 + 2$ mA (top) and $1 + 3$ mA (bottom): envelope excursion, quadratic excursion, mean-square polarization, and isotropic beat. The first three use the fixed 1:1 envelope-maximizing direction. Each color scale ends at its 1:1 gray-matter 99th percentile. Circles and triangles mark the projected centroids of the top 10% of ROI nodes at 1:1 and 1:3, not whole-brain maxima. *(e)* Target-to-reference contrast; dashed curves use square-root maps to match degree. *(f)* Anterior focus displacement: top 10% (line), range over 5–50% (band). *(g)* Contrasts at 1:1, raw (solid) and degree-matched (hatched). The reference is gray matter outside the ROI. All maps precede detector gain: $\mathcal{I}$ weights $Y_{\mathrm{DC}} = \sigma'' \mathcal{I}/2$, but does not set its sign.

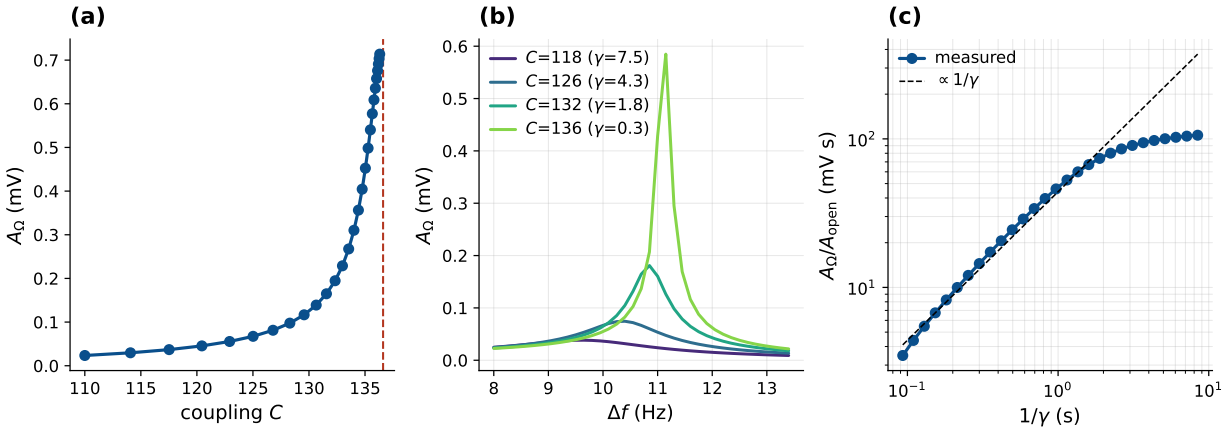


Figure 10: Coupling controls the amplification at fixed detection (Jansen–Rit). The input is the AM surrogate of Eq. (17). The field-free operating point v^* is pinned by co-solving $p(C)$, so $\sigma''(v^*) = -0.151\ \mathrm{s}^{-1}\mathrm{mV}^{-2}$ and A_{open} are held constant throughout. *(a)* The demodulated A_Ω grows as the coupling C approaches the Hopf C^* from the stable-focus side ($\gamma > 0$). *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain $A_\Omega/A_{\mathrm{open}}$, in mV s, follows $1/\gamma$ (dashed) and saturates near the bifurcation.

vantage. For the AM surrogate with amplitude $\varepsilon_{\mathrm{TI}}$ and modulation depth m, and a tACS input of amplitude $\varepsilon_{\mathrm{tACS}}$, the ratio of voltage lock-in amplitudes in linear response is

$$\mathcal{R}_{\mathrm{lin}} = \frac{|\sigma''(v^*)|\,\varepsilon_{\mathrm{TI}}^2 m}{2\sigma'(v^*)\,\varepsilon_{\mathrm{tACS}}}.$$

The common network transfer cancels, so the ratio has no dependence on γ when both responses are linear about the same state. At the amplitudes used in Figs. 10 and S3, $\mathcal{R}_{\mathrm{lin}} = 0.143$. Genuine two tones give output ratios of 0.14 far from the Hopf, 0.24 at $\gamma = 1.07\ \mathrm{s}^{-1}$, and 0.46 at $0.12\ \mathrm{s}^{-1}$. The AM-surrogate comparison is reported in the supplement (§S2.3). These are response ratios for the stated unequal input amplitudes, not an equal-RMS comparison. The equal-per-carrier and equal-total-peak comparisons with both tACS and tDCS are given in §2.6. In controls at three couplings, matching the tACS input to TI's leading beat source gives two-tone TI/tACS output ratios of 0.99, 0.93, and 0.90. The stronger direct response saturates earlier near the Hopf: multiplying its input by 6.99 increases the output by only 6.97, 3.95, and 1.96 at these three couplings. Stimulation-induced mean shifts, which field-free pinning does not cancel, account for this departure from the linear ratio.

4.8 Exact mean-field confirmation: the NMM2 PING gamma generator

The exact PING mean field generates and amplifies the beat through its derived v_E^2 nonlinearity. With a 300 Hz two-tone field entering the excitatory current, the lock-in of r_E shows a growing gamma resonance below the Hopf and a response superposed on autonomous gamma above it (Figs. 11, S8). The resonance ridge intensifies through the Hopf; its appreciable frequency shift with operating point makes it diagonal, unlike the JR sweep shown here [33].

This derived detector lies on the fast side of its crossover at 300 Hz: carrier voltage excursions mix through v_E^2 before the rate reads the result. For the full eight-state PING (Eq. (19)), each high-frequency current tone of amplitude I_j produces a voltage phasor $I_j/(i\omega_j\tau_E)$. The v_E^2 term therefore supplies a difference-frequency source of magnitude $I_1 I_2/(\tau_E^2\omega_1\omega_2)$ inside the equation for $\tau_E\dot{v}_E$. After division by τ_E, this source is propagated through the network susceptibility at Ω. At four stable operating points, the resulting response agrees with the full second-order calculation to within 1.1% at 300 Hz, 0.1% at 1 kHz, and 0.03% at 2 kHz (Fig. 12b).

In NMM2 PING, J is the mean inter-neuron QIF synaptic weight. Sweeping J at fixed background

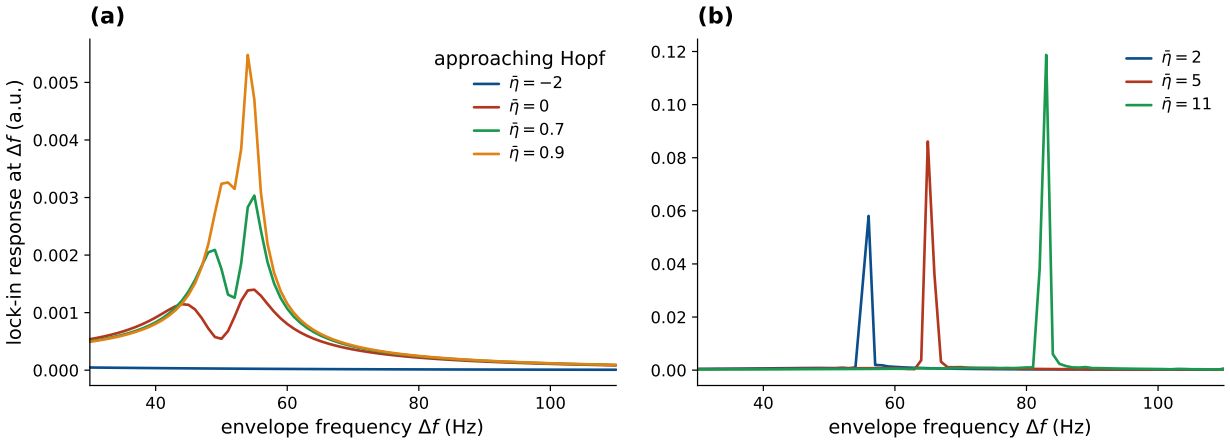


Figure 11: NMM2 PING gamma resonance (exact mean field, genuine two tones). Lock-in response of the excitatory rate r_E at the envelope frequency Δf under a two-tone TI field (carrier 300 Hz). *(a)* (stable focus, $\bar{\eta} < \bar{\eta}_H = 1.110$) forced resonance at the gamma frequency ($f_0 \approx 54$ Hz), the peak growing $0.0014 \to 0.0030 \to 0.0055$ ms^{-1} as the Hopf is approached ($1/\gamma$ gain). *(b)* (limit cycle, $\bar{\eta} > \bar{\eta}_H$) response in the autonomous-gamma regime, the peak shifting upward with $\bar{\eta}$ ($56 \to 65 \to 83$ Hz). The input carries no power at Δf; the response is synthesized by the exact quadratic v^2 nonlinearity, with no sigmoid anywhere in the model. AM/two-tone comparisons are reported in the supplement (§S2.3).

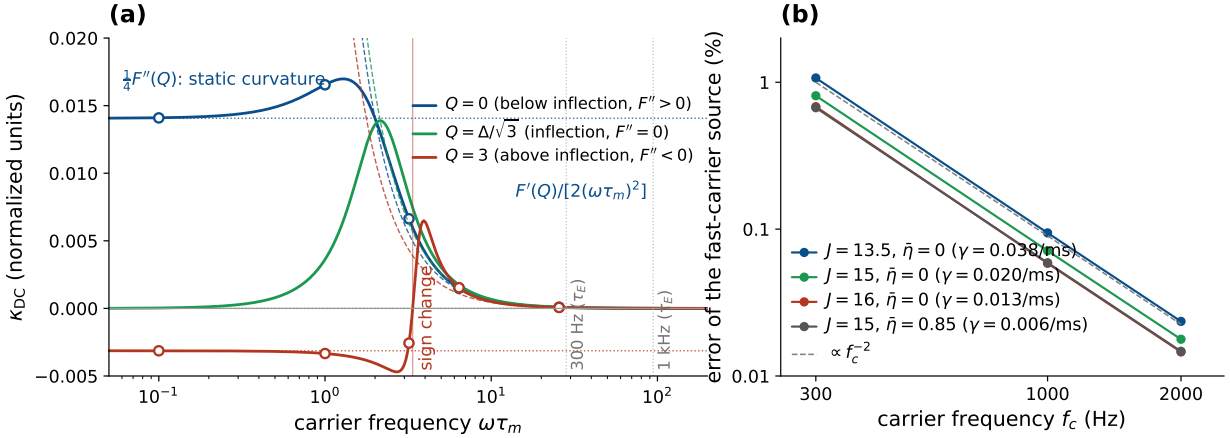


Figure 12: Static curvature at slow carriers, slope at fast ones. *(a)* Second-order DC rate shift of an uncoupled MPR population ($\Delta = 1$) under a current carrier, Eq. (B6), for operating points below, at and above the inflection of $F(Q)$. Dotted: the static limit $\frac{1}{4}F''$; dashed: the fast-carrier limit $F'/[2(\omega\tau_m)^2]$; circles: RK4 integrations at $I_{ac} = 0.05$ averaged over whole carrier periods (relative error below 10^{-3}). Above the inflection the sign changes with carrier frequency. The horizontal coordinate is $\omega\tau_m$; the carrier markers use $\tau_m = \tau_E = 15$ ms. *(b)* PING of Eq. (19): relative error of the fast-carrier difference-frequency source against the exact second-order response at four stable operating points (γ the damping of the gamma mode), two tones centered at 300, 1000 and 2000 Hz.

drive toward the gamma Hopf reveals two competing modes (Fig. S10). A second mode near 55 Hz carries the linear-response peak over $14.6 \lesssim J \lesssim 16.2$; the critical mode carries it at lower couplings and again near the Hopf, where its frequency is about 50 Hz. In that near-Hopf interval, $0.0014 \lesssim \gamma \lesssim 0.0081$ ms^{-1}, the linear gain scales as $1/\gamma$ with a fitted exponent of 1.02 over a 5.8-fold damping range. The full-sweep exponent is 0.71, so a single-mode law does not describe the whole curve. Coupling-dependent sensitivity [33] therefore includes changes in which mode carries the peak.

At $A_f = 8$, the simulated response exceeds the normalized linear reference by up to 48% in the amplitude check, then saturates. At a quarter of that amplitude ($A_f = 2$, one sixteenth of the

power), the departures stay within 11% at the nine tested couplings. The driven endpoint remains attracting under both waveforms (supplement, §S4.3). The alpha JR and gamma NMM2 circuits share the $1/\gamma$ gain scaling of their critical mode despite their different mixers, with amplification adjustable through coupling.

4.9 QIF responses distinguish beat modulation from field-induced gamma

Below the field-induced Hopf, the QIF network shows beat-locked modulation with a small mean-rate change; above it, the field induces gamma and adds a beat component. The 8000-neuron populations test this distinction at the spike level [31, 27, 33] (Figs. 13, 14).

An unmodulated carrier matched in RMS input current (model units) provides this control in the mean field (Fig. 14c,d). It produces a rectified E-drive shift within 3% of TI's and a similar mean-rate increase. At $A_f = 35$, both waveforms induce gamma near 51 Hz, whereas TI also produces the 55 Hz beat component. At $A_f = 10$, neither induces gamma, and TI alone produces a resolved beat response. A longer window reduces the single carrier's apparent 55 Hz depth from 0.011 to 9.2×10^{-5}, while TI's stays at 0.30–0.31; the former is spectral leakage. The equal-RMS control separates induction of gamma from the beat response and has been tested only in the mean field.

The $A_f = 35$ runs are finite-amplitude illustrations: the fast-carrier voltage excursion is about 1.2 per tone in model units. At $A_f = 10$, the estimated rectified E-drive shift is 0.12, below the E-only Hopf threshold $\delta\eta_E^* \approx 0.286$; the stronger drive exceeds it, as quantified below.

At $\bar{\eta} = 0.85$ and $A_f = 10$, beat depths span about 0.09–0.16 across cutoff and population choices, versus 0.13 in the mean field. The cutoff-free theta form gives 0.156 at $N = 8000$ and 0.137 at $N = 16000$, a 12% change exceeding the pilot tolerance. The baseline mean-rate increase is 2.8%, but the modulation's size is not converged (supplement, §S4.5). At $\bar{\eta} = 1.5$, the original finite-cutoff run gives a 42 Hz depth of about 0.02 and a 1.6% mean-rate increase; ongoing gamma dominates.

At the larger amplitude $A_f = 35$ and $\bar{\eta} = 0.85$, stimulation produces a dominant line near 52 Hz, with modulation depth about 0.70, and a smaller component at the 55 Hz beat, with depth 0.29. The second-order carrier-averaged system predicts an excitatory drive shift $\delta\eta_E \approx 1.65$, beyond the E-only Hopf threshold $\delta\eta_E^* \approx 0.286$, with a leading eigenvalue whose real part is $+0.032$ ms^{-1}. Direct mean-field integration reproduces the off-beat line, supporting the interpretation of panel (b) as field-induced gamma with a smaller forced component.

At $\bar{\eta} = 1.5$, gamma activity is already present without stimulation. With the field on, the rhythm near 53–54 Hz remains about four times larger than the 42 Hz beat-locked component. Gamma therefore dominates the raster in panel (d); the coexistence of the rhythms produces slower variations in burst height. Folding the rate on the beat phase suppresses the non-beat-locked gamma and exposes the smaller component. The inset shows that phase average, not the full rate variation.

The beat and ongoing gamma mix, as in endogenous amplitude modulation [25], producing sidebands near 95 and 11 Hz (Fig. S12b), establishing nonlinear mixing without distinguishing gamma-amplitude gating from phase modulation; field-free gamma frequencies agree between the spiking network and mean field (panel a).

At $A_f = 35$, the mean rate rises by about 8–11% across the two conditions, while the beat modulation depth, $A_\Omega/\langle r_E \rangle$, rises from below 0.003 to 0.29 ($\bar{\eta} = 0.85$) and 0.23 ($\bar{\eta} = 1.5$) (Fig. S12c). Modulation is about three times the relative mean-rate change, but per-neuron locking is weak: the pairwise phase consistency of single-neuron spike phases at Δf rises from its field-free control (median -0.02) to a median of 0.003–0.005, and the pooled vector strength from below 0.02 to 0.12–0.13. The population's beat-locked component is built from many weakly biased cells; these finite-amplitude runs do not establish its size under weak physiological stimulation.

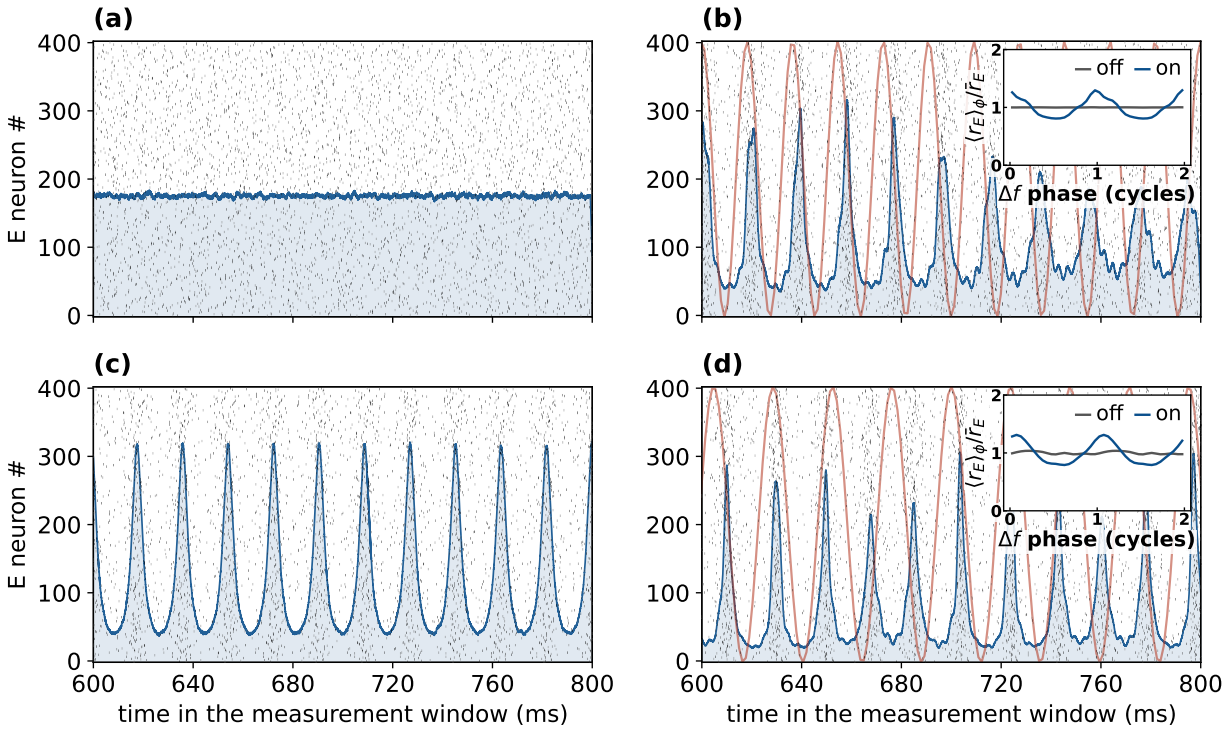


Figure 13: Beat-locked activity superposed on gamma. QIF network, $N = 8000$ per population, $A_f = 35$ per tone, carrier 300 Hz. Black ticks: 400 E neurons; blue: smoothed rate on its own scale; red: envelope in arbitrary units. The last 200 ms are shown. *(a,b)* Field off/on at $\bar{\eta} = 0.85$, beat 55 Hz: induced gamma near 52 Hz dominates. *(c,d)* Field off/on at $\bar{\eta} = 1.5$, beat 42 Hz: autonomous gamma remains dominant. Insets fold rate on beat phase over 44 and 34 cycles, respectively, normalized to each condition's mean (off gray, on blue; scale 0–2). Folding suppresses non-beat-locked gamma and exposes the smaller beat component; it does not establish gamma-amplitude gating or phase capture.

5 Discussion

5.1 Cellular access and mixing require separate calibration

The JR and QIF models separate mixing from recurrent amplification, but neither calibrates the full pathway from physiological kHz fields to a neural response. The short-time cases in Table 2 are access hypotheses with preparation-dependent coupling; changing the QIF time constant tests sensitivity to those assumptions. Cellular mixing [2, 10] and network amplification [16, 17, 6] supply complementary evidence for the stages, not a joint calibration.

At fields of order 1 V/m, channels respond through membrane polarization, well below fields associated with direct protein effects [42].[2] Terminals can polarize more strongly than somata [43, 44, 45]. Small size alone does not imply fast charging: specific membrane resistance times capacitance is area-independent; conductance, geometry, and stimulation mode determine access. The supplementary comparison distinguishes these mechanisms from strong-pulse charging (§S1).

Missey et al. [46] report biomarker suppression with both low- and high-kHz carriers. Their Methods specify 1300/1430 Hz at one center and 9000/9130 Hz at the other centers. For these pairs, the 0.2 ms point-membrane model predicts a high/low source ratio of 0.030 at matched local fields. This comparison challenges that access hypothesis but does not test it directly: carrier band was confounded with center, local polarization was unmeasured, and a nonlinear or saturating physiological readout could reconcile different sources with similar biomarker changes.

[2] That review describes direct effects at 10 kV/m to GV/m, depending on molecule and protocol; it excludes fields below 0.1 kV/m, where it expects direct effects to be negligible against thermal noise.

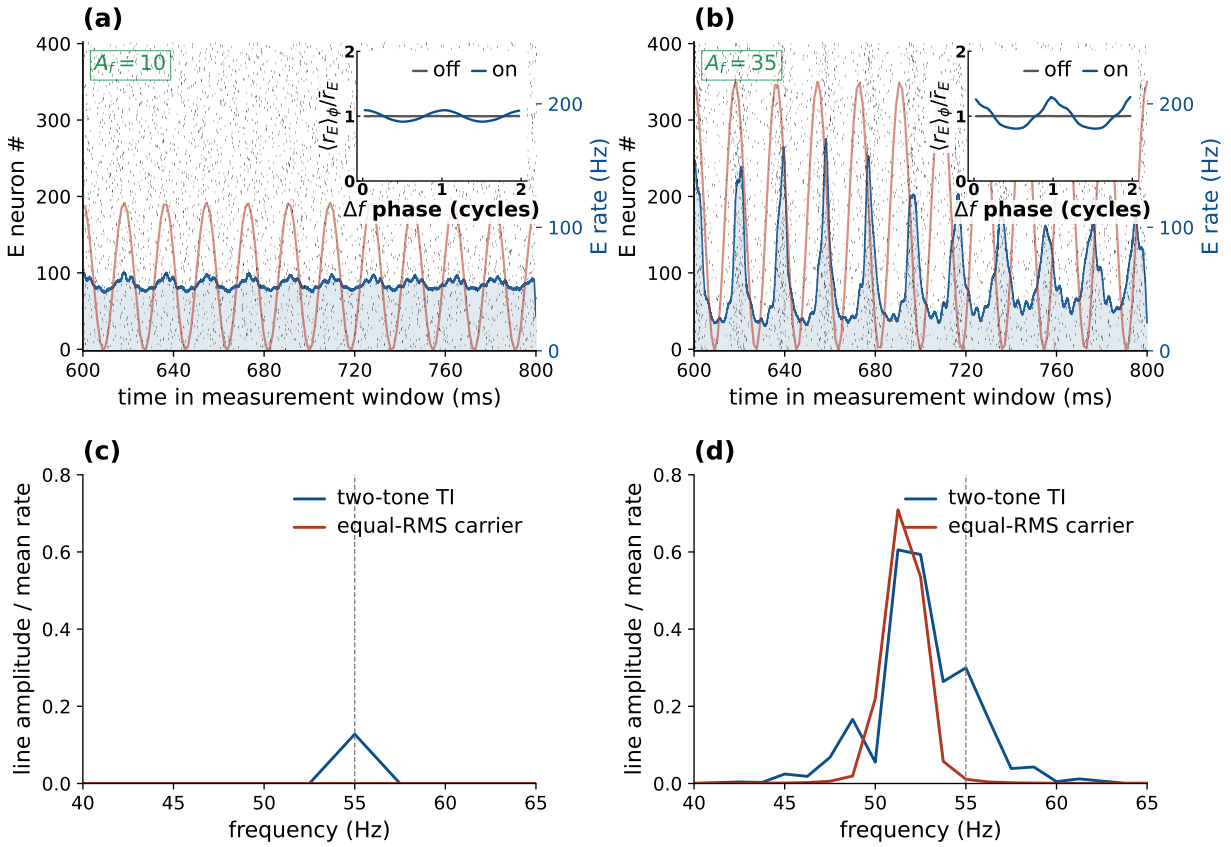


Figure 14: TI and an equal-RMS carrier across the gamma threshold. Background $\bar{\eta} = 0.85$, carrier 300 Hz, beat 55 Hz. *(a,b)* Spiking network at $A_f = 10$ and 35 per tone. Black: 400 of 8000 E neurons; blue: population rate on a common scale; red: envelope. Insets show mean-normalized phase folds (off gray, on blue). Beat depths at this cutoff are about 0.09 and 0.3, with mean-rate increases of 2.8% and 10.5%; the larger input also induces gamma. *(c,d)* Mean-field spectra under TI (blue) and one carrier of peak amplitude $\sqrt{2}A_f$ (red), normalized to mean rate. Both induce gamma at the larger input; TI adds the 55 Hz beat (dashed). Hann windows span 800 ms after 800 ms settling. Spiking panels use $V_{\text{peak}} = 100$; weak-drive depth depends on cutoff and population size. RMS matching is of input current in model units, not physical power.

5.2 Detector laws predict different spatial responses

The head-model comparison separates geometric consequences of detector laws. The product and summed-squares maps differ from the envelope map even before state-dependent detection and network gain are applied. The fixed quadratic pattern follows algebraically from fixed field shapes and linear access; opposite envelope and mean-square centroid motion is a result for the tested subject and montages, not a general steering law. A sharp rectifier can follow the envelope over a finite amplitude range while smoothing or heterogeneity restores quadratic behavior at smaller inputs. Amplitude and harmonic sweeps therefore complement the spatial comparison.

The connection to Violante et al. [37] is a hypothesis generated by these maps. Their models and cadaver recordings supported anterior envelope steering from $2 + 2$ mA to $1 + 3$ mA. During participants' 32 s encoding blocks, 1:3 reduced the evoked blood-oxygen-level-dependent (BOLD) response relative to sham ($P = 0.006$) and 1:1 ($P = 0.015$); 1:1 did not differ from sham ($P = 0.953$). The stimulation-by-region interaction was nonsignificant ($P = 0.76$), so the physiology did not establish spatial steering. Significant voxelwise reductions extended beyond the anterior focus: 30% of anterior, 25% of posterior, and 2% of middle hippocampal voxels.

In IXI002, mean-square polarization rises by 24% under 1:3, while mean envelope excursion falls by 45% and both quadratic beat maps fall by 25%. At fixed uniform static curvature the steady-

source magnitude therefore rises by 24%; a mean shift from rectification of the beat instead scales as $(3/4)^2 \approx 0.56$ in the weak-response limit. The first ordering is consistent with stronger BOLD suppression if the steady source lowers activity and the hemodynamic response preserves that ordering. A slow linear readout suppresses a zero-mean 5 Hz beat while retaining a shift over the stimulation window; nonlinear neural or vascular responses can also convert that beat into a mean change or alter the encoding response.

This comparison does not identify the biological detector. Local envelope excursion can increase under 1:3 where $|\mathcal{P}_1| > 2|\mathcal{P}_2|$, reaching 1.5× where $|\mathcal{P}_1| \geq 3|\mathcal{P}_2|$ (supplement, §S3.5); a selective or thresholded envelope response remains possible. Equal summed currents do not match local RMS polarization, and the nonsignificant 1:1–sham result does not establish a null effect. Heterogeneous curvature and unknown recurrent and hemodynamic gains can change the predicted ordering. Measuring spatially resolved steady and beat responses at controlled polarization and state, with an equal-RMS carrier control, would discriminate the hypotheses.

5.3 Field-induced oscillations and resonant amplification

At stronger fields, TI and unmodulated carriers can have similar excitation thresholds [21, 23], and peripheral responses need not extract the envelope [22]. Our weak-input calculation concerns modulation of ongoing activity [20]; the finite-amplitude QIF examples instead show a mean shift crossing a circuit Hopf. That boundary depends on amplitude and initial state, and supplies no physiological threshold. The supplementary laminar extension tests two resonant bands at low carriers (§S6), outside TI's kHz regime.

The common TI/tACS amplifier follows from matching source, readout, and state, consistent with excitatory–inhibitory models [15, 33] and related tACS/AM phase-locking studies [47, 48, 49]. Stimulation-induced state shifts can change the finite-amplitude ratio. Near-Hopf temporal slowing does not by itself establish a diverging spatial correlation length. Writing the reverberation time as $\tau_{\mathrm{rev}} = 1/\gamma$, the quality factor is $\mathcal{Q} = \pi f_0 \tau_{\mathrm{rev}}$ and power bandwidth approximately $1/(\pi\tau_{\mathrm{rev}})$. Relaxation over a few hundred milliseconds [50] gives order-ten quality and order-1 Hz bandwidth at 10 Hz, making detuning consequential; that measured relaxation is not specific to an alpha mode.

5.4 Efficiency depends on the generated drive

Measured DC couplings are 0.12–0.18 mV per V/m in hippocampus (0.21 at 10 Hz) [51, 52] and reach 0.5 mV per V/m in cortical L5/6 pyramidal cells [39]; human cortical fields near 0.8 V/m have been reported at 2 mA [53, 54]. These measurements constrain illustrative input budgets, not the kHz transfer. The 2–5% source comparison in §2.6 matches each TI carrier's polarization to the direct input. Matching the total TI peak instead halves the ratio to $\mathcal{R}_{\mathrm{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$; common recurrent gain cancels in linear response.

For $E_j = 0.5$ V/m, $\lambda_0 = 0.5$ mV per V/m, $f_c = 2$ kHz and the assumed $\tau_m = 0.2$ ms, access gives $\varepsilon \approx 0.092$ mV. Rounding to 0.1 mV yields $\mathcal{R}_{\mathrm{det}} \approx 0.8\%$ at $v^* = 8.30$ mV under total-peak matching. This is an assumed input budget, not an inferred polarization in recorded cells.

For comparison, the carrier-averaged, matched-peak, open-loop ratio has a numerically estimated supremum of 0.348 over bounded nondecreasing memoryless transfers (supplement, §S5, Eq. (S7)). A positive mixture of step responses extends the logistic calculation to this class. Approaching the bound requires a transition sharp relative to the input: microvolt logistic widths at the illustrative polarizations. Its step-like limit has a second-harmonic ratio near 0.15, differing from weak square-law and linear envelope detection. The bound excludes dynamic QIF and general cellular mechanisms. Increasing a common matched polarization cannot exceed it; unequal polarizations invalidate the matching. Unequal-current TI and matched-depth AM-tACS comparisons do not measure this open-loop ratio.

Vieira et al.'s weaker TI effects [20] are compatible with a smaller beat source, but their local mixer polarizations were unmatched (supplement, §S5.1). Phase-locking value (PLV) also depends on baseline and state. A calibrated amplitude sweep with a common readout is needed to estimate the detector coefficient.

5.5 Limits of the population description

Synaptic filters before a detector further attenuate its input. Model choice also matters: Hodgkin–Huxley and Frankenhaeuser–Huxley models reproduce stimulus-evoked firing missed by the tested leaky, exponential, and adaptive-exponential integrate-and-fire models [55]. Noise, additional heterogeneity, and intercolumn interactions can broaden or shift our deterministic resonances. Human findings include peripheral effects without central effects [56] and null occipito-parietal alpha-power modulation [57]. Access, generated sources, and circuit tuning therefore need measurement in the same preparation.

5.6 Falsifiable predictions and tests

Table 4 separates tests of access, detection, and amplification from observations already consistent with the model. Quantitative reanalysis of the timing and synaptic-blockade data requires the amplitude and harmonic controls of §3.2, with detector and network changes measured separately.

Network filtering must be separated from the harmonic signature. At resonance with $\mathcal{Q} = 10$, a detector ratio of 20% appears as only 0.7%, which could make an envelope follower look quadratic. Driving off resonance at $\Delta f \simeq f_0/4$ places both harmonics on the flatter part of the response, where $|G(2\Omega)/G(\Omega)| \approx 1.22$–$1.25$ for $\mathcal{Q} = 3$–30. Alternatively, a weak direct tACS sweep can measure G independently so its effect can be removed from the harmonic ratio.

Model prediction	Discriminating measurement
Access and detector dynamics	Sweep carrier frequency and tonic state at fixed beat. Measure local polarization and network gain separately; normalize the beat source by the carrier product and compare its magnitude and phase.
Quadratic versus envelope detection	Vary carrier amplitudes independently, then together. A weak quadratic source follows their product; an envelope follower follows the minimum. Measure amplitude scaling and harmonics off resonance or correct their network gains; a single harmonic ratio is insufficient.
Fixed quadratic spatial pattern	Vary the current ratio at fixed electrodes, frequencies, and state. Compare normalized physiological beat maps with fixed local orientations. Use a cortex-only reference for deep selectivity; measure steady shifts and repeat at lower amplitude to test state changes.
Resonant amplification	Sweep the beat through the intrinsic band across states, measuring peak frequency, height, and width. A matched direct tACS source tests the shared amplifier.
Population modulation and mean response	Compare rate lock-in and mean shift in the same units. Synaptic blockade tests recurrent amplification alongside an independent cellular mixing measurement. Distinguish rate modulation from single-neuron phase locking.
Steady versus beat-dependent effects	Compare TI with an unmodulated carrier of matched local RMS input and duration, checking the DC coefficient across the carrier interval. Measure mean activity, intrinsic oscillations, and the beat separately.

Table 4: Tests separating the proposed mechanisms. The detailed evidence matrix and qualifications are in the supplement (§S7.3). RMS matching refers to the detector input, not electrode or tissue power.

6 Conclusion

TI requires access, detection, and amplification. For weak inputs with a nonzero second-order coefficient, detection generates a beat proportional to the product of the local carrier polarizations and a steady source proportional to their summed squares. Fast axonal access can be represented by a short effective membrane time, with its transfer still requiring calibration.

The detector law changes the spatial prediction. In the single-subject geometric illustration, changing the current ratio moves the hippocampal envelope and mean-square centroids in opposite directions but only rescales the quadratic beat at fixed neural state. Under uniform scalar coupling and detection in homogeneous conductivity, the steady-source magnitude has no strict interior maximum.

Near a Hopf bifurcation, the circuit amplifies the beat in inverse proportion to modal damping. TI shares this linear amplifier with matched direct low-frequency stimulation. In the pinned JR sweep, the response is expressed mainly as beat-locked population-rate modulation; its ratio to the mean shift eventually falls as the oscillation rectifies. At larger fields, the steady shift alone can induce gamma, as can an equal-RMS unmodulated carrier.

Local-polarization measurements, independent-amplitude tests, beat sweeps across states, and equal-RMS controls separate these mechanisms. Together they test whether a measured TI response reflects access, nonlinear detection, recurrent gain, or a field-induced change of state.

Code and data availability

Simulation code, cached results, and figure-generation scripts are provided at the paper's GitHub repository; the public archive is linked through Zenodo (doi:10.5281/zenodo.21009618). The repository `README` maps figures to their scripts and distinguishes recomputation from replotting cached data. The two-tone kHz example and both LaNMM carrier-measurement windows have cached outputs and reproduction scripts. The FEM producer `code/fem_quadmaps.py` computes the reported maps and statistics from the stored fields; the summary arrays and `code/make_fem_fig.py` reproduce Fig. 9 without the head model. Recomputing its maps from the underlying fields currently requires Neuroelectrics' nonpublic TI library and the original IXI002 fields and labels. A NumPy-only consumer, `code/fem_portable.py`, and a checked exporter are included; the per-node export and its comparison with the published statistics remain pending. Input hashes, settings, and export instructions are recorded in `docs/fem_ixi002_violante.md`, `code/fem_quadmaps.json`, and `docs/fem_portable.md`. The code in this paper's repository is released under CC-BY-4.0.

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Competing interests

Giulio Ruffini, Ricardo Salvador, Borja Mercadal, and Francesca Castaldo are employees of Neuroelectrics, which develops and sells transcranial electrical stimulation devices. Giulio Ruffini is a co-founder and shareholder of Neuroelectrics. The other authors declare no competing interests.

Author contributions

Using the CRediT taxonomy: **G. Ruffini**: conceptualization, methodology, formal analysis, software, investigation, visualization, writing (original draft), supervision, funding acquisition. **R. Sal-**

vador: conceptualization (current-ratio steering), methodology (head modeling), software, formal analysis, data curation, visualization. **B. Mercadal**: validation, visualization, writing (review and editing). **R. de Palma Aristides**: software, visualization, writing (review and editing). **F. Castaldo**: conceptualization, writing (review and editing). **N. Grossman**: conceptualization. **S. Canals**: supervision, writing (review and editing). **C. Mirasso**: supervision, writing (review and editing).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude and OpenAI ChatGPT to streamline the narrative and to assist with code and figure generation from data the authors produced. The authors reviewed and edited all resulting content and take full responsibility for the content of the published article.

References

[1] Nir Grossman, David Bono, Nina Dedic, Suhasa B. Kodandaramaiah, Andrii Rudenko, Ho-Jun Suk, Antonino M. Cassara, Esra Neufeld, Niels Kuster, Li-Huei Tsai, Alvaro Pascual-Leone, and Edward S. Boyden. Noninvasive deep brain stimulation via temporally interfering electric fields. *Cell*, 169(6):1029–1041.e16, 2017.

[2] Ehsan Mirzakhalili, Beatrice Barra, Marco Capogrosso, and Scott F. Lempka. Biophysics of temporal interference stimulation. *Cell Systems*, 11(6):557–572.e5, 2020.

[3] Paul J. Nahin. *The Science of Radio: with MATLAB and Electronics Workbench Demonstrations*. Springer, 2nd edition, 2001.

[4] Yair Deitcher, Guy Eyal, Lida Kanari, Matthijs B. Verhoog, Guy Antoine Atenekeng Kahou, Huibert D. Mansvelder, Christiaan P. J. de Kock, and Idan Segev. Comprehensive morpho-electrotonic analysis shows 2 distinct classes of L2 and L3 pyramidal neurons in human temporal cortex. *Cerebral Cortex*, 27(11):5398–5414, 2017.

[5] Guy Eyal, Matthijs B. Verhoog, Guilherme Testa-Silva, Yair Deitcher, Johannes C. Lodder, Ruth Benavides-Piccione, Juan Morales, Javier DeFelipe, Christiaan P. J. de Kock, Huibert D. Mansvelder, and Idan Segev. Unique membrane properties and enhanced signal processing in human neocortical neurons. *eLife*, 5:e16553, 2016.

[6] Zeinab Esmaeilpour, Greg Kronberg, Davide Reato, Lucas C. Parra, and Marom Bikson. Temporal interference stimulation targets deep brain regions by modulating neural oscillations. *Brain Stimulation*, 14(1):55–65, 2021.

[7] Charles C. H. Cohen, Marko A. Popovic, Jan Klooster, Marie-Theres Weil, Wiebke Möbius, Klaus-Armin Nave, and Maarten H. P. Kole. Saltatory conduction along myelinated axons involves a periaxonal nanocircuit. *Cell*, 180(2):311–322.e15, 2020.

[8] Maarten H. P. Kole, Susanne U. Ilschner, Björn M. Kampa, Stephen R. Williams, Peter C. Ruben, and Greg J. Stuart. Action potential generation requires a high sodium channel density in the axon initial segment. *Nature Neuroscience*, 11(2):178–186, 2008.

[9] Boshuo Wang, Aman S. Aberra, Warren M. Grill, and Angel V. Peterchev. Responses of model cortical neurons to temporal interference stimulation and related transcranial alternating current stimulation modalities. *Journal of Neural Engineering*, 19(6):066047, 2022.

[10] Charlotte E. Luff, Robert Peach, Emma-Jane Mallas, Edward Rhodes, Felix Laumann, Edward S. Boyden, David J. Sharp, Mauricio Barahona, and Nir Grossman. The neuron mixer and its impact on human brain dynamics. *Cell Reports*, 43(6):114274, 2024.

[11] Bruce Hutcheon and Yosef Yarom. Resonance, oscillation and the intrinsic frequency preferences of neurons. *Trends in Neurosciences*, 23(5):216–222, 2000.

[12] Raul de Palma Aristides, Leonardo L. Gollo, Francesca Castaldo, Giulio Ruffini, Santiago Canals, and Claudio R. Mirasso. Frequency-selective responses to temporal interference stimulation in hippocampal neurons. Unpublished manuscript, September 29, 2026. BCOM Working Paper WP0242, 2026.

[13] Nicolas Brunel and Vincent Hakim. Fast global oscillations in networks of integrate-and-fire neurons with low firing rates. *Neural Computation*, 11(7):1621–1671, 1999.

[14] Nicolas Brunel and Xiao-Jing Wang. What determines the frequency of fast network oscillations with irregular neural discharges? I. synaptic dynamics and excitation–inhibition balance. *Journal of Neurophysiology*, 90(1):415–430, 2003.

[15] Erwan Ledoux and Nicolas Brunel. Dynamics of networks of excitatory and inhibitory neurons in response to time-dependent inputs. *Frontiers in Computational Neuroscience*, 5, 2011.

[16] Sara Caldas-Martinez, Chaitanya Goswami, Mats Forssell, Jiaming Cao, Alison L. Barth, and Pulkit Grover. Cell-specific effects of temporal interference stimulation on cortical function. *Communications Biology*, 7:1076, 2024.

[17] Nafiseh Karimi, Rassoul Amirfattahi, and Abolghasem Zeidaabadi Nezhad. Neuromodulation effect of temporal interference stimulation based on network computational model. *Frontiers in Human Neuroscience*, 18, September 2024.

[18] Hossein Soroushi, Sara Abbasi, Yun Du, Ning Ning, and Yuming Lei. Temporal interference stimulation: mechanisms, optimization, validation, and clinical prospects—a comprehensive review. *WIREs Computational Statistics*, 17(3):e70031, 2025.

[19] Wenjiang Wang, Florian Missey, Chenxi Ling, Kun Wei, Junhao Jian, Haoxuan Sun, Liujiao Yan, Shaohua Hu, and Bo Bi. Temporal interference stimulation: a new paradigm for non-invasive deep brain stimulation—mechanistic insights, clinical advances, and future directions. *Translational Psychiatry*, 2026.

[20] Pedro G. Vieira, Matthew R. Krause, and Christopher C. Pack. Temporal interference stimulation disrupts spike timing in the primate brain. *Nature Communications*, 15:4558, 2024.

[21] Aleksandar Opančar, Petra Ondráčková, David Samuel Rose, Jan Trajlinek, Vedran Đerek, and Eric Daniel Głowacki. The same biophysical mechanism is involved in both temporal interference and direct kHz stimulation of peripheral nerves. *Nature Communications*, 16:9006, 2025.

[22] Ryan B. Budde, Michael T. Williams, and Pedro P. Irazoqui. Temporal interference current stimulation in peripheral nerves is not driven by envelope extraction. *Journal of Neural Engineering*, 20(2):026041, 2023.

[23] Iurii Semenov, Vitalii Kim, Giedre Silkuniene, and Andrei G. Pakhomov. Excitation and polarization of isolated neurons by high-frequency sine waves for temporal interference stimulation. *Cell Reports Physical Science*, 6(7):102660, 2025.

[24] Giulio Ruffini, Edmundo Lopez-Sola, Raul Palma, Roser Sanchez-Todo, Jakub Vohryzek, Francesca Castaldo, and Karl Friston. Cross-frequency coupling as a neural substrate for prediction error evaluation: a laminar neural mass modeling approach. *Neural Computation*, 38(8):1299–1343, 2026.

[25] Giulio Ruffini. Neural encoding through hierarchical amplitude modulation. *bioRxiv*, 2025. Preprint 2025.11.03.686310.

[26] Antonino M. Cassarà, Taylor H. Newton, Katie Zhuang, Sabine J. Regel, Peter Achermann, Alvaro Pascual-Leone, Niels Kuster, and Esra Neufeld. Recommendations for the safe application of temporal interference stimulation in the human brain part II: Biophysics, dosimetry, and safety recommendations. *Bioelectromagnetics*, 46(1):e22536, 2025.

[27] Francesca Castaldo, Raul de Palma Aristides, Pau Clusella, Jordi Garcia-Ojalvo, and Giulio Ruffini. Rosetta Stone of neural mass models. *Physics Reports*, 1189:1–49, 2026. arXiv:2512.10982.

[28] Ben H. Jansen and Vincent G. Rit. Electroencephalogram and visual evoked potential generation in a mathematical model of coupled cortical columns. *Biological Cybernetics*, 73(4):357–366, 1995.

[29] Hugh R. Wilson and Jack D. Cowan. Excitatory and inhibitory interactions in localized populations of model neurons. *Biophysical Journal*, 12(1):1–24, 1972.

[30] Adrià Galan-Gadea, Ricardo Salvador, Fabrice Bartolomei, Fabrice Wendling, and Giulio Ruffini. Spherical harmonics representation of the steady-state membrane potential shift induced by tDCS in realistic neuron models. *Journal of Neural Engineering*, 20(2):026004, 2023.

[31] Ernest Montbrió, Diego Pazó, and Alex Roxin. Macroscopic description for networks of spiking neurons. *Physical Review X*, 5(2):021028, 2015.

[32] Giulio Ruffini. Analysis and extension of exact mean-field theory with dynamic synaptic currents. *bioRxiv*, 2021. Preprint 2021.09.01.458563 (NMM2).

[33] Pau Clusella, Elif Köksal-Ersöz, Jordi Garcia-Ojalvo, and Giulio Ruffini. Comparison between an exact and a heuristic neural mass model with second-order synapses. *Biological Cybernetics*, 117(1-2):5–19, 2023.

[34] Arkady Pikovsky, Michael Rosenblum, and Jürgen Kurths. *Synchronization: A Universal Concept in Nonlinear Sciences*. Cambridge University Press, 2001.

[35] Ryan T. Canolty and Robert T. Knight. The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11):506–515, 2010.

[36] Adriano B. L. Tort, Robert Komorowski, Howard Eichenbaum, and Nancy Kopell. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2):1195–1210, 2010.

[37] Ines R. Violante, Ketevan Alania, Antonino M. Cassarà, Esra Neufeld, Emma Acerbo, Romain Carron, Adam Williamson, Danielle L. Kurtin, Edward Rhodes, Adam Hampshire, Niels Kuster, Edward S. Boyden, Alvaro Pascual-Leone, and Nir Grossman. Non-invasive temporal interference electrical stimulation of the human hippocampus. *Nature Neuroscience*, 26(11):1994–2004, 2023.

[38] Ricardo Salvador, Junhong Zhou, Brad Manor, and Giulio Ruffini. Group optimization methods for dose planning in tES. *Journal of Neural Engineering*, 22:046045, 2025.

[39] Thomas Radman, Raddy L. Ramos, Joshua C. Brumberg, and Marom Bikson. Role of cortical cell type and morphology in subthreshold and suprathreshold uniform electric field stimulation in vitro. *Brain Stimulation*, 2(4):215–228.e3, October 2009.

[40] François Grimbert and Olivier Faugeras. Bifurcation analysis of Jansen's neural mass model. *Neural Computation*, 18(12):3052–3068, 2006.

[41] Martin Vinck, Marijn van Wingerden, Thilo Womelsdorf, Pascal Fries, and Cyriel M. A. Pennartz. The pairwise phase consistency: a bias-free measure of rhythmic neuronal synchronization. *NeuroImage*, 51(1):112–122, 2010.

[42] Michal Cifra, Saurabh Kumar Pandey, Tomáš Zakar, Michaela Poplová, Daniela G. Blanco Campoy, Jiří Průša, Daniel Havelka, Paolo Marracino, Micaela Liberti, Francesca Apollonio, Niall J. English, Carl Caleman, and Erik G. Marklund. The effects of external electric fields on proteins. *Chemical Society Reviews*, 55(14):7568–7616, 2026.

[43] Asif Rahman, Davide Reato, Mattia Arlotti, Fernando Gasca, Abhishek Datta, Lucas C. Parra, and Marom Bikson. Cellular effects of acute direct current stimulation: somatic and synaptic terminal effects. *The Journal of Physiology*, 591(10):2563–2578, 2013.

[44] Aman S. Aberra, Ruochen Wang, Warren M. Grill, and Angel V. Peterchev. Multi-scale model of axonal and dendritic polarization by transcranial direct current stimulation in realistic head geometry. *Brain Stimulation*, 16(6):1776–1791, 2023.

[45] Darpan Chakraborty, Dennis Q. Truong, Marom Bikson, and Hanoch Kaphzan. Neuromodulation of axon terminals. *Cerebral Cortex*, 28(8):2786–2794, 2018.

[46] Florian Missey, Emma Acerbo, Adam S. Dickey, Jan Trajlinek, Ondřej Studnička, Claudia Lubrano, Mariane de Araújo e Silva, Evan Brady, Vit Všianský, Johanna Szabo, Irena Dolezalova, Daniel Fabo, Martin Pail, Claire-Anne Gutekunst, Rosanna Migliore, Michele Migliore, Stanislas Lagarde, Romain Carron, Fariba Karimi, Raul Castillo Astorga, Antonino M. Cassara, Niels Kuster, Esra Neufeld, Fabrice Bartolomei, Nigel P. Pedersen, Robert E. Gross, Viktor Jirsa, Daniel L. Drane, Milan Brázdil, and Adam Williamson. Non-invasive temporal interference stimulation of the hippocampus suppresses epileptic biomarkers in patients with epilepsy: biophysical differences between kilohertz and amplitude modulated stimulation. *Brain Stimulation*, 19(1):102981, 2026.

[47] M. M. Ali, K. K. Sellers, and F. Frohlich. Transcranial alternating current stimulation modulates large-scale cortical network activity by network resonance. *Journal of Neuroscience*, 33(27):11262–11275, July 2013.

[48] Ehsan Negahbani, Florian H. Kasten, Christoph S. Herrmann, and Flavio Fröhlich. Targeting alpha-band oscillations in a cortical model with amplitude-modulated high-frequency transcranial electric stimulation. *NeuroImage*, 173:3–12, June 2018.

[49] Matthew R. Krause, Pedro G. Vieira, and Christopher C. Pack. Transcranial electrical stimulation: How can a simple conductor orchestrate complex brain activity? *PLOS Biology*, 21(1):e3001973, January 2023.

[50] Jens Wilting and Viola Priesemann. Inferring collective dynamical states from widely unobserved systems. *Nature Communications*, 9(1), June 2018.

[51] Marom Bikson, Masashi Inoue, Hiroki Akiyama, Jackie K. Deans, John E. Fox, Hiroyoshi Miyakawa, and John G. R. Jefferys. Effects of uniform extracellular dc electric fields on excitability in rat hippocampal slices in vitro. *The Journal of Physiology*, 557(1):175–190, May 2004.

[52] Jacqueline K. Deans, Andrew D. Powell, and John G. R. Jefferys. Sensitivity of coherent oscillations in rat hippocampus to ac electric fields. *The Journal of Physiology*, 583(2):555–565, August 2007.

[53] Yu Huang, Anli A Liu, Belen Lafon, Daniel Friedman, Michael Dayan, Xiuyuan Wang, Marom Bikson, Werner K Doyle, Orrin Devinsky, and Lucas C Parra. Measurements and models of electric fields in the in vivo human brain during transcranial electric stimulation. *eLife*, 6, February 2017.

[54] Yu Huang, Anli A Liu, Belen Lafon, Daniel Friedman, Michael Dayan, Xiuyuan Wang, Marom Bikson, Werner K Doyle, Orrin Devinsky, and Lucas C Parra. Correction: measurements and

models of electric fields in the in vivo human brain during transcranial electric stimulation. *eLife*, 7, February 2018.

[55] Tom Plovie, Ruben Schoeters, Thomas Tarnaud, Wout Joseph, and Emmeric Tanghe. Nonlinearities and timescales in neural models of temporal interference stimulation. *Bioelectromagnetics*, 46(1):e22522, 2025.

[56] Krisztián Iszak, Simon Mathies Gronemann, Stefanie Meyer, Alexander Hunold, Jana Zschüntzsch, Mathias Bähr, Walter Paulus, and Andrea Antal. Why temporal inference stimulation may fail in the human brain: A pilot research study. *Biomedicines*, 11(7):1813, June 2023.

[57] Paria Mansourinezhad, Rob M C Mestrom, Debby C W Klooster, Mathieu Sprengers, Paul A J M Boon, and Margarethus M Paulides. Systematic review of experimental studies in humans on transcranial temporal interference stimulation. *Journal of Neural Engineering*, 22(5):051001, September 2025.

A Notation and units

Table 5 groups the symbols by their role in the calculation. JR and LaNMM voltages are in mV and rates in s^{-1}. QIF voltages and current-like drives are dimensionless model variables; QIF rates are physical rates, integrated in ms^{-1} and converted to Hz where indicated. The uncoupled MPR derivation explicitly uses membrane-time units; the main population equations retain physical time and rate. All frequencies and time constants in a formula must use the same time unit. In the FEM comparison $\mathbf{n}^*$ is the direction maximizing the envelope at balanced currents, held fixed during the ratio sweep. A star otherwise marks a fixed point or, on a bifurcation parameter, its critical value. Subscripts E and I identify excitatory and inhibitory populations; they are not field or current variables.

Table 5: Notation and units. "Normalized" means dimensionless in the stated model scaling; output units depend on the named observable. Equation numbers prefixed by S identify supplementary definitions; B-prefixed numbers identify article-appendix equations, and unprefixed numbers identify main-text equations. Lettered locations identify article appendices.

Symbol	Meaning	Units	Defined in
Field and access			
$E(t), E_0, \mathbf{E}_j$	Applied field, its amplitude, and vector field of tone $j = 1, 2$	V/m	Eq. (3)
$s(t), \varepsilon_j, \varepsilon$	Local polarization; tone amplitudes; common amplitude ($\varepsilon_{\mathrm{TI}}$ and $\varepsilon_{\mathrm{tACS}}$ label the waveform)	mV	Eqs. (1), (4)
$\varepsilon_{\mathrm{D}}, \varepsilon_{\mathrm{A}}, \varepsilon_{\mathrm{eq}}$	Direct DC polarization, signed AC amplitude, and equivalent direct polarization matching a TI source	mV	§2.6
$I_j^{\mathrm{pair}}, \mathcal{P}_j$	Peak electrode-pair current; signed local polarization per unit current	mA; mV/mA	§2.3
$\mathbf{F}_j, \sigma_{\mathrm{cond}}$	FEM field per pair current; tissue conductivity	(V/m)/mA; S/m	§3.3
$\mathcal{A}(t), \overline{\mathcal{A}}$	Nonnegative polarization envelope and its beat-period mean	mV	Eq. (8); Table 1
f_1, f_2, f_c	Tone frequencies and their center	Hz	Eq. (1)
ω_j, ω_c	Angular frequencies $2\pi f_j, 2\pi f_c$	rad/s	Eqs. (1), (4)
$\Delta f, \Omega$	Beat frequency $f_2 - f_1$; $\Omega = 2\pi\Delta f$	Hz; rad/s	Eq. (1)
m	AM-surrogate modulation depth	dimensionless	Eq. (17)
$\Lambda[E], \lambda_0, \lambda_f$	Field-induced polarization; static and frequency-dependent coupling	mV; mV/(V/m)	Eqs. (B1), (4)
$\boldsymbol{\lambda}_0^k, \boldsymbol{\lambda}_f^k$	Coupling vectors for compartment k	mV/(V/m)	Eq. (4)
H, H_k, H_m	Dominant-compartment access transfer, indexed transfer, and point-membrane transfer	dimensionless	Fig. 2; §2.2, §2.8
τ_m, τ	Membrane time; access-filter time in kHz controls	ms	Eqs. (15), (13)
f_{char}	Characteristic frequency $1/(2\pi\tau)$	Hz	Table 2
$\ell(\omega), \ell_0$	Frequency-dependent and static cable space constants	μm	App. S1.1, Eq. (S1)
$\mathbf{V}, \mathsf{C}_m, \mathsf{G}_m$	Compartment voltages, capacitance and conductance matrices	V; F; S	App. S1.1
$\mathbf{I}^0, \mathbf{p}_E$	Endogenous currents; field-to-current vector	A; A/(V/m)	App. S1.1
$\mathbf{x}, \phi_j, \mathbf{n}$	Position, field potential, unit cell orientation	m; V; unit vector	§2.3, §2.5
$\mathcal{I}$	Nonnegative mean-square local polarization before detector gain	mV^2	§2.5
Detection			
$\sigma, \sigma', \sigma''$	JR/LaNMM rate curve, slope and curvature	s^{-1}; $s^{-1}mV^{-1}$; $s^{-1}mV^{-2}$	Eq. (2)

Continued on next page

Table 5 continued

Symbol	Meaning	Units	Defined in
e_0, v_0, ρ	Half the rate ceiling, inflection voltage, inverse voltage scale	s^{-1}; mV; mV^{-1}	Eq. (2)
PEIX	Signed sigmoid saturation index	dimensionless	Eq. (14)
Y_Ω, Y_{DC}	Complex beat amplitude and signed steady source before recurrence	detector-output units	Eqs. (6), (5)
a_j^k, b_j^k, B_{kl}	Compartment polarization, polarization per current, bilinear mixing coefficient	mV; mV/mA; output/mV^2	§2.3
$\chi^{(2)}$	Dynamic coefficient mapping a polarization product to a rate beat	$s^{-1}mV^{-2}$	§2.2
g, T, Y_{LF}	Memoryless response, envelope transduction, carrier-averaged output	specified output units	Eq. (10)
c_1, c_2, c_4	Linear, quadratic and quartic envelope coefficients	output/$(mV)^{1,2,4}$	§2.4
$\mathcal{H}_{det}, \mathcal{H}_{meas}$	Second/fundamental harmonic magnitude ratios before and after the network	dimensionless	Eq. (S3)
$\mathcal{M}[y]$	Peak-to-peak excursion of the quantity y	units of y	Eq. (9)
$F(Q), F', F''$	MPR steady-rate curve and drive derivatives	normalized	Eq. (16)
Q, q	Mean total drive $\bar{\eta} + I$; single-cell drive $\eta + I$	normalized	Eq. (16); App. B.3
I_{ac}	Single-tone current amplitude	normalized	§2.10
κ_{DC}, κ_V	Mean-rate mixing per squared current or squared voltage excursion	normalized	Eqs. (B6), (S2)
d, ν, ω_z	MPR relaxation, ringing and zero-crossing angular frequencies	membrane-time units	App. B.2
$\mathsf{J}, \mathbf{h}, H_v$	Reduced MPR Jacobian, unit-current response and voltage transfer	normalized	§2.10; App. B.3
$\mathbf{b}_2, \mathsf{I}_2$	Quadratic DC source per I_{ac}^2; identity matrix	normalized	App. B.3
$\mathcal{R}_{det}, \mathcal{R}_{lin}$	TI/direct-drive source ratio; linear TI/tACS output ratio	dimensionless	App. S5; §4.7
δ, ζ	Threshold offset; sigmoid rate divided by $2e_0$	mV; dimensionless	§2.4; App. S5
u, ξ, β, k_{ell}^2	Reduced amplitude $\rho\varepsilon$, offset $\rho(v^* - v_0)$, threshold ratio, elliptic parameter	dimensionless	App. S5
$L, \psi, \mathsf{K}, \mathsf{E}$	Unit logistic, normalized waveform, complete elliptic integrals	dimensionless	App. S5
Network			
r_X, v_X	Population firing rate and mean potential, indexed by X	s^{-1}; mV in JR/LaNMM, normalized in QIF	§2.1
t, r, f (MPR derivation)	Time, population rate and single-cell rate in the stated membrane-time convention	normalized locally	App. B.2
$r_{syn}, r_E^{syn}, r_I^{syn}$	Synaptically filtered firing rates	same rate units as input	Eqs. (B5), (19)
$u_{PSP}, u_{X\leftarrow Y}, \hat{L}$	Postsynaptic potential and inverse synaptic operator	mV; maps mV to s^{-1}	Eq. (B1)
A, B, a, b	JR excitatory/inhibitory PSP gains and inverse synaptic times	mV; s^{-1}	§2.1; App. B
h_e, h_i	JR excitatory and inhibitory synaptic impulse responses	mV	Fig. 2
$C, C_{1\ldots4}, C_{X\leftarrow Y}$	JR connectivity scale and pathway weights	dimensionless	§2.1; App. B
y_0, y_1, y_2, v	JR PSPs; voltage readout $v = y_1 - y_2$	mV	App. B
p, p_H, C^*	JR external rate and its upper Hopf value; critical connectivity	s^{-1}; dimensionless for C^*	§3.4, §4.5
$\eta_j, \bar{\eta}, \Delta$	QIF excitabilities, Lorentzian center and half-width	normalized	Eq. (15)

Continued on next page

Table 5 continued

Symbol	Meaning	Units	Defined in
$I, I_E, I_j, A_f, \lambda_I$	QIF input, E-population input, tone amplitudes, forcing amplitude and field coupling	normalized; λ_I in $(\mathrm{V/m})^{-1}$	Eqs. (15), (19)
$\mathcal{U}_j$	Total synaptic and field drive to QIF neuron j	normalized	Fig. 2
J, J^*, w_{XY}	QIF coupling, critical coupling and relative pathway weights	dimensionless	Eq. (19)
$\tau_E, \tau_I, \tau_a, \tau_g, \tau_s$	E/I membrane and excitatory/inhibitory/generic synaptic times	ms	Eqs. (B5), (19)
$\hat{K}_a, \hat{K}_g, H_s$	QIF synaptic rate filters; JR rate-to-voltage transfer	dimensionless; mV s for H_s	§3.5, §2.7
$\mu_\pm, \gamma, \omega_0, f_0$	Modal eigenvalues, damping, angular and ordinary modal frequency	s^{-1} (ms^{-1} where stated); Hz for f_0	Eq. (11)
$\tau_{\mathrm{rel}}, \tau_{\mathrm{rev}}, \mathcal{Q}$	Population relaxation time, $1/\gamma$, and mode quality factor	time; dimensionless for $\mathcal{Q}$	§2.1, §2.7, §5
D, G, R_Ω	Beat-source coefficient, recurrent transfer, complex observed beat response	input/output-dependent	Eq. (12)
θ_j	QIF phase with $V_j = \tan(\theta_j/2)$	rad	§S4.5
$V_j, V_{\mathrm{peak}}, N$	Single-QIF voltage, spike cutoff and neurons per population	normalized; count	§3.6
$\delta\eta_E, \delta\eta_E^*$	Rectified E-drive shift and its Hopf threshold	normalized	§3.6, §4.9
P_1, P_2, r_P, φ_X	Deep/superficial pyramidal populations; JR and LaNMM pyramidal rates	labels; s^{-1} for rates	§4.6; §S6.1
$z, \alpha, \mathcal{F}, \varepsilon_{\mathrm{SL}}$	Stuart–Landau mode, growth rate, forcing and forcing amplitude	normalized mode; s^{-1} for the others	App. S2.4
Measurement			
$L_{\cos}, L_{\sin}, A_\Omega$	Signed lock-in quadratures and nonnegative amplitude	measured voltage or rate units	Eq. (18)
A_{open}	Magnitude of the open-loop rate beat	s^{-1}	§4.5
$w_k, t_k, v_k, \bar{v}$	Hann weights, sample times, voltage samples and weighted mean	1; s; mV; mV	Eq. (18)
$A_{\mathrm{test}}, \theta_0$	Amplitude and phase of the lock-in test tone	readout units; rad	§3.2
$\Delta t, N_{\mathrm{grid}}$	Integration step and number of parameter combinations	s or ms; count	§3.4
$A_\Omega/\langle r_E \rangle$	Beat-locked population modulation depth	dimensionless	§3.6
PPC	Pairwise phase consistency of single-neuron spike phases	dimensionless	§3.6
$f_{\mathrm{out}}, S(f_c)$	Dominant output frequency; LaNMM mean off-alpha lock-in	Hz; mV	App. S7; §S6.1

B Simulated models: equations and parameters

For a synapse from population Y to X, the resulting first-generation neural mass model (NMM1) equations are [27]

$$\hat{L}_{X\leftarrow Y}[u_{X\leftarrow Y}] = C_{X\leftarrow Y}\, r_Y, \qquad v_X = \Lambda[E] + \sum_{Y:\, C_{X\leftarrow Y}\neq 0} u_{X\leftarrow Y}, \qquad r_X = \sigma(v_X). \tag{B1}$$

The operator $\hat{L}$ is the inverse synaptic filter, $u_{X\leftarrow Y}$ its postsynaptic potential, $C_{X\leftarrow Y}$ the connectivity, and $\Lambda[E]$ the direct polarization caused by the applied field E (V/m). The rates r_X, r_Y are in s^{-1}, the potentials $v_X, u_{X\leftarrow Y}$ in mV, and the pathway weights $C_{X\leftarrow Y}$ dimensionless. This field contribution enters alongside the synaptic potentials, without first passing through a synapse.

Synaptic operators convert rates into postsynaptic potentials, which sum with the direct field polarization before the rate curve is applied.

Jansen–Rit column

Writing y_0 for the excitatory PSP driving the interneurons, y_1 and y_2 for the excitatory and inhibitory PSPs onto the pyramidal population, external input p, and the applied-field perturbation $s(t)$ (§2.2) entering the pyramidal sigmoid,

$$\ddot{y}_0 = Aa\,\sigma\big(y_1 - y_2 + s(t)\big) - 2a\dot{y}_0 - a^2 y_0, \tag{B2}$$

$$\ddot{y}_1 = Aa\,\big[\,p + C_2\,\sigma(C_1 y_0)\,\big] - 2a\dot{y}_1 - a^2 y_1, \tag{B3}$$

$$\ddot{y}_2 = Bb\,C_4\,\sigma(C_3 y_0) - 2b\dot{y}_2 - b^2 y_2, \tag{B4}$$

The pyramidal output passes through h_e to give y_0; the excitatory and inhibitory interneurons receive $C_1 y_0$ and $C_3 y_0$, and return through weights C_2 and C_4. The excitatory and inhibitory impulse responses are $h_e(t) = Aa\,te^{-at}$ and $h_i(t) = Bb\,te^{-bt}$ for $t \geq 0$. The equations are integrated as six first-order ODEs; the LFP proxy is $v = y_1 - y_2$ [28]. Table 6 gives the reference column parameters. The synaptic rates correspond to excitatory and inhibitory decay times of 10 and 20 ms, while A and B set the postsynaptic-potential gains. The sigmoid parameters fix its voltage scale and rate ceiling; the sweeps vary the external rate p or connectivity C to change the operating state, as specified in Methods.

Table 6: Jansen–Rit parameters [28].

Symbol	Meaning	Value
A, B	excit./inhib. PSP gain	3.25, 22 mV
a, b	synaptic rates	100, 50 s^{-1}
v_0, e_0, ρ	sigmoid inflection, max rate/2, slope	6 mV, 2.5 s^{-1}, 0.56 mV^{-1}
C	global connectivity	135
C_1, C_2, C_3, C_4	couplings	C, $0.8C$, $0.25C$, $0.25C$
p	external input (bifurcation knob)	0–400 Hz

The alpha limit cycle lies between Hopf points near $p = 90$ and 316 s^{-1}. It coexists with a lower stable fixed point up to the saddle-node on an invariant circle (SNIC, $p \approx 114$), then with a large-amplitude cycle up to its saddle-node near $p = 137$. The SNIC terminates the large-amplitude branch, not the alpha cycle.

B.1 Static rate with dynamic synapses

A second-order synapse with time constant τ_s obeys

$$r(t) \simeq \frac{F(Q(t))}{\tau_m}, \qquad \tau_s^2 \ddot{r}_{\mathrm{syn}} + 2\tau_s \dot{r}_{\mathrm{syn}} + r_{\mathrm{syn}} = r(t), \tag{B5}$$

where r_{syn} is the filtered rate entering the recurrent drive in Q, with the original coupling normalization retained. This is the slow-synapse limit: shortening the effective membrane time makes the transfer appear static while the network retains its synaptic dynamics.

B.2 Static and dynamic MPR mixing coefficients

In this subsection and the derivation that follows, time is measured in units of τ_m, rate in units of $1/\tau_m$, and ω denotes angular frequency in membrane-time units. Thus $r^* = F(Q)$ and physical angular frequency is ω/τ_m. Consider an uncoupled MPR population at drive Q, with a weak current $I_{\mathrm{ac}} \cos \omega t$. To second order in I_{ac}, the carrier-averaged shift of r is $I_{\mathrm{ac}}^2 \kappa_{\mathrm{DC}}(Q, \omega)$, where

$$\kappa_{\mathrm{DC}}(Q,\omega) = \frac{r^* \left[\omega^2 + 3d^2 - \nu^2\right]}{(d^2+\nu^2)\{[d^2+\nu^2-\omega^2]^2 + 4d^2\omega^2\}}, \qquad r^* = F(Q), \quad d = \frac{\Delta}{\pi r^*}, \quad \nu = 2\pi r^*. \tag{B6}$$

The physical rate shift is $I_{\mathrm{ac}}^2 \kappa_{\mathrm{DC}}/\tau_m$. The fixed-point eigenvalues in normalized time are $-d \pm i\nu$: d is the relaxation rate and ν the ringing angular frequency, both in membrane-time units; their physical values are d/τ_m and ν/τ_m. Expansion about this fixed point gives the limits

$$\kappa_{\mathrm{DC}}(Q,\omega) \xrightarrow[\omega \to 0]{} \tfrac{1}{4} F''(Q), \qquad \kappa_{\mathrm{DC}}(Q,\omega) \xrightarrow[\omega \gg d,\nu]{} \frac{F'(Q)}{2\omega^2}. \tag{B7}$$

In the first limit the population tracks its static curve, whose curvature rectifies the input. In the second, the voltage excursion is approximately I_{ac}/ω; its square supplies a slow drive that the population reads through F'. Above the inflection of F, these limits have opposite signs and the coefficient vanishes at $\omega_z = \sqrt{\nu^2 - 3d^2}$. The 300 Hz and kHz carriers are on the fast side for the parameters of the pyramidal–interneuron network gamma (PING) circuit studied here. That classification is parameter-dependent: a shorter effective membrane time can bring the same physical carrier into the adiabatic regime.

B.3 Single-cell rate and dynamic MPR mixing

The MPR steady-rate curve F of §2.10, Eq. (16), averages the single-neuron rate $f(q) = \sqrt{q}/\pi$ for $q > 0$, and $f(q) = 0$ otherwise, over the excitability distribution. Here η is the neuron's excitability and $q = \eta + I$ its constant total drive.

The coefficient in Eq. (B6) follows by expanding about the fixed point (r^*, v^*), with $v^* = -d/2$, in membrane-time units. The Jacobian and unit-current phasor are

$$\mathsf{J} = \begin{pmatrix} 2v^* & 2r^* \\ -2\pi^2 r^* & 2v^* \end{pmatrix}, \qquad \mathbf{h} = (i\omega \mathsf{I}_2 - \mathsf{J})^{-1} \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

where I_2 is the identity matrix and the first-order response is $I_{\mathrm{ac}} \operatorname{Re}(\mathbf{h} e^{i\omega t})$. Writing h_r, h_v for its rate and voltage components, with an overbar denoting complex conjugation and Re the real part, the carrier-averaged quadratic source per I_{ac}^2 is

$$\mathbf{b}_2 = \begin{pmatrix} \operatorname{Re}(h_r \overline{h_v}) \\ (|h_v|^2 - \pi^2 |h_r|^2)/2 \end{pmatrix}.$$

The mean second-order correction is $-I_{\mathrm{ac}}^2 \mathsf{J}^{-1} \mathbf{b}_2$; its rate component gives Eq. (B6).